

Initiating Search

November 10, 2025, 9:10 PM

• Search:

CAS Lexicon Search:

Ugi reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (2,268)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (1,622)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	
Language:	English	
Publication Year:	2020 to 2025	

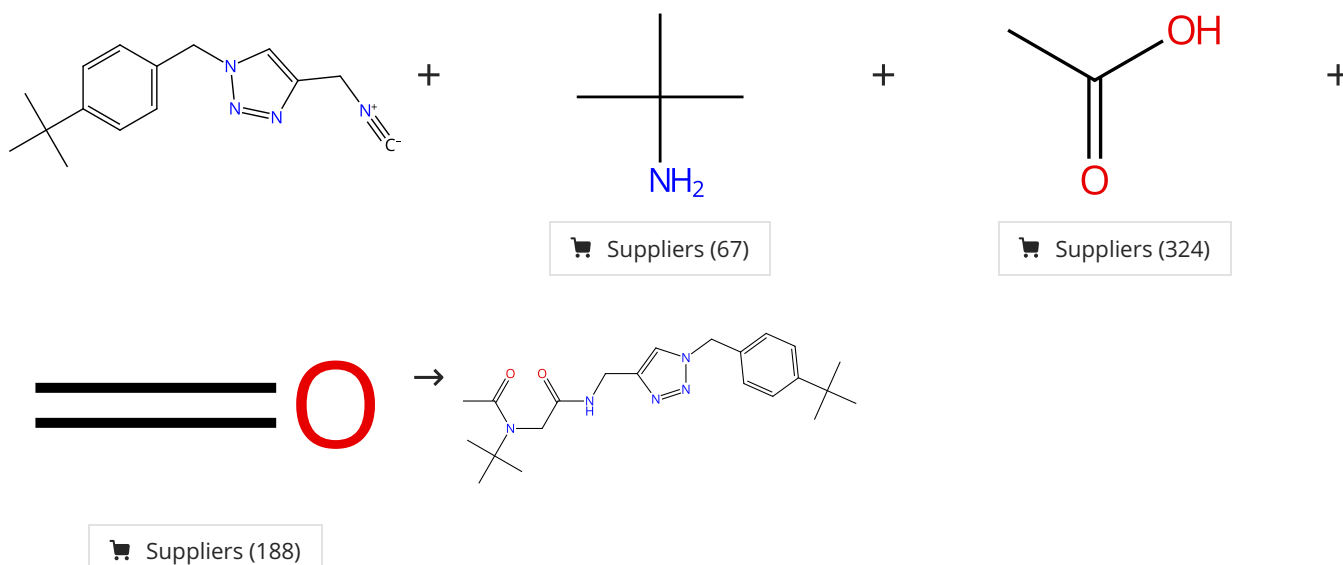
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 Reactions (50)[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-36691960

Steps: 1 Yield: 100%

Synthesis, Characterization and Chemistry of Tetrakis (Propargylisocyanide) Copper(I) Complex

1.1 Solvents: Methanol, Water; 0 °C; 18.5 h, 28 °C

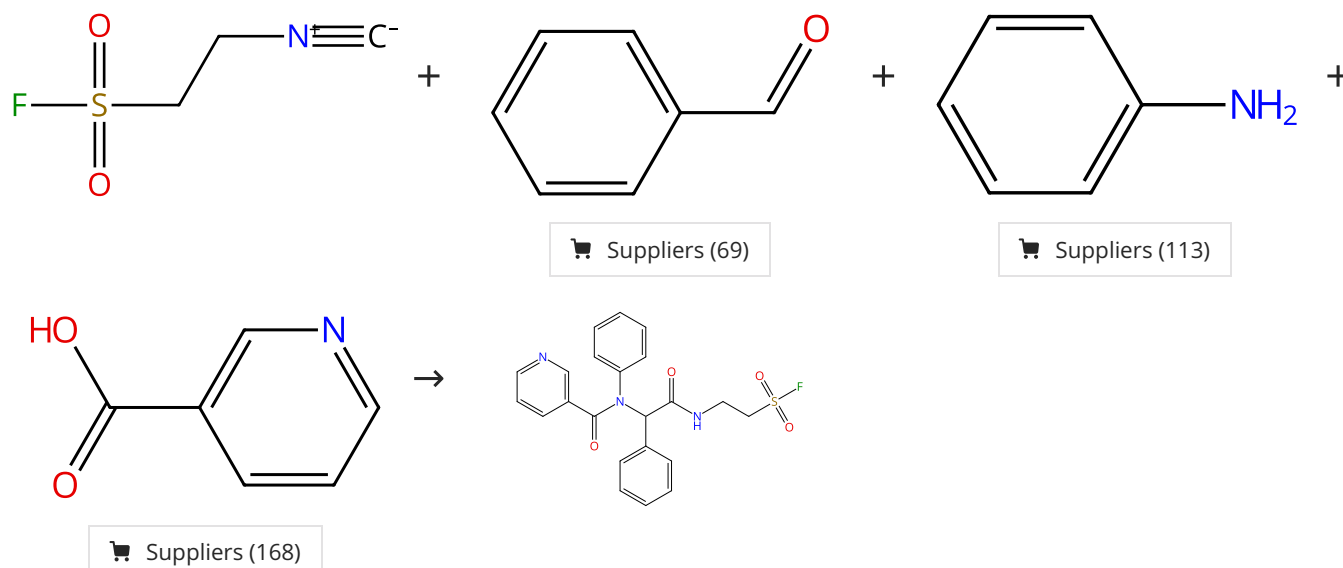
Experimental Protocols

By: Quelhas, Alexandre; et al

Molbank (2023), 2023(1), M1599.

Scheme 2 (1 Reaction)

Steps: 1 Yield: 99%



31-367-CAS-23904445

Steps: 1 Yield: 99%

SuFExable isocyanides for Ugi reaction: Synthesis of sulfonyl fluoro peptides

1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 3 h, rt

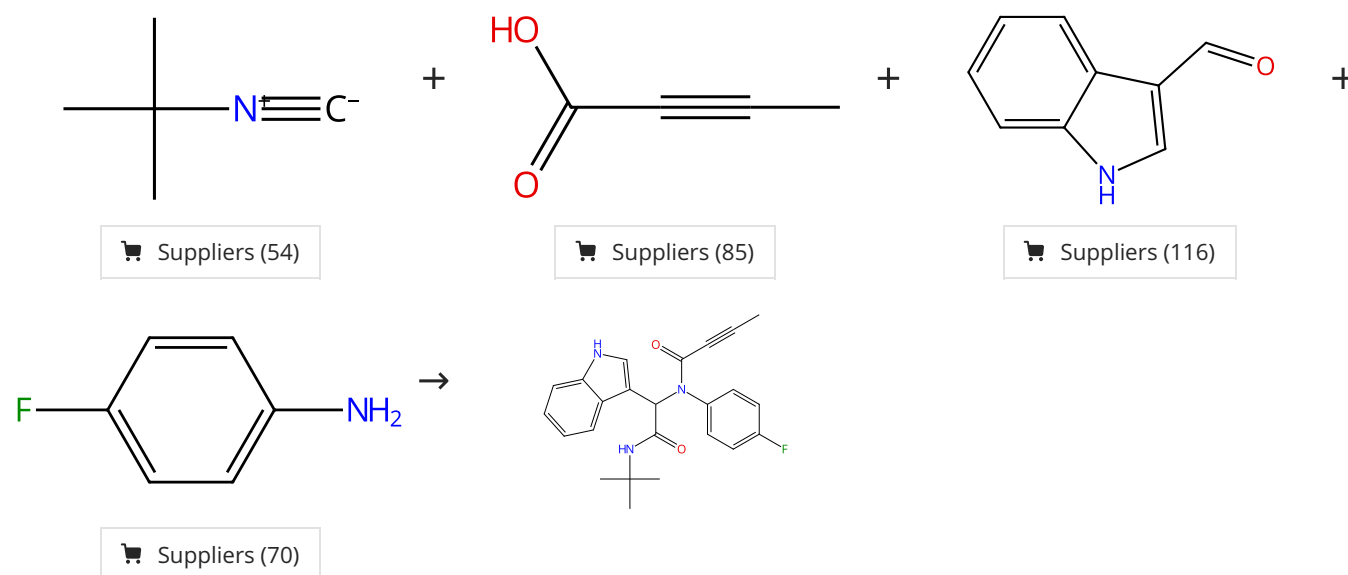
Experimental Protocols

By: Xu, Shuheng; et al

Organic Letters (2021), 23(13), 5197-5202.

Scheme 3 (1 Reaction)

Steps: 1 Yield: 99%



31-367-CAS-21006755

Steps: 1 Yield: 99%

Silver(I) Triflate-Catalyzed Protocol for the Post-Ugi Synthesis of Spiroindolines

1.1 Solvents: Methanol; 24 h, rt

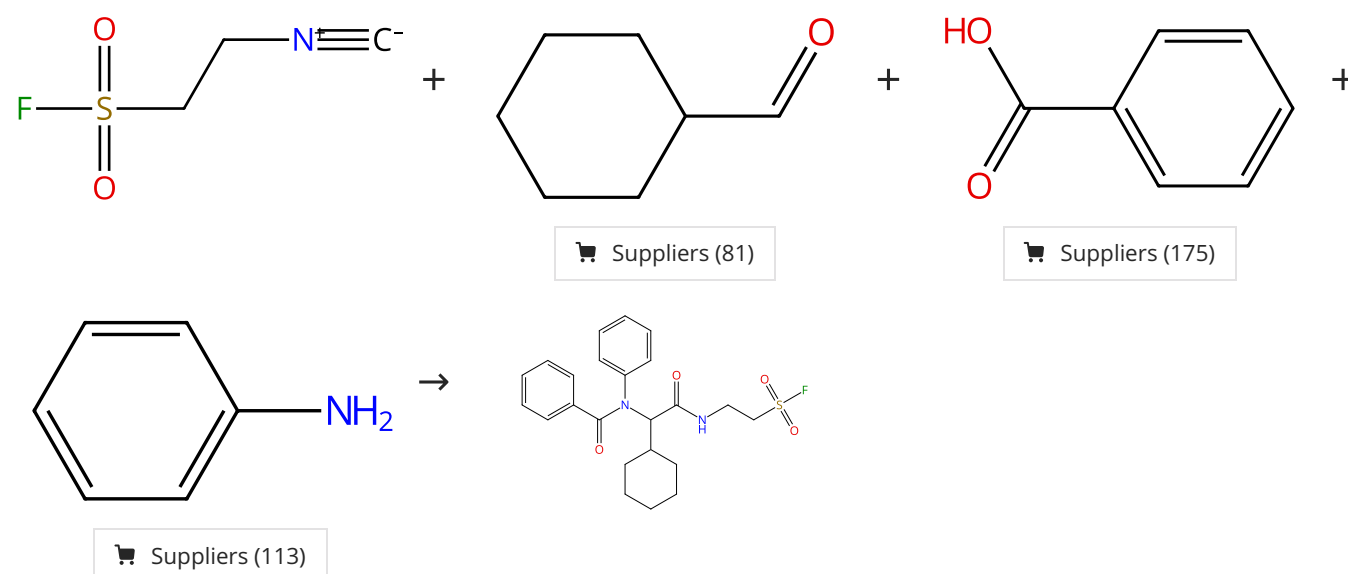
By: Zaman, Manzoor; et al

Experimental Protocols

Advanced Synthesis & Catalysis (2020), 362(1), 261-268.

Scheme 4 (1 Reaction)

Steps: 1 Yield: 99%



31-367-CAS-23902642

Steps: 1 Yield: 99%

SuFExable isocyanides for Ugi reaction: Synthesis of sulfonyl fluoro peptides

1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 3 h, rt

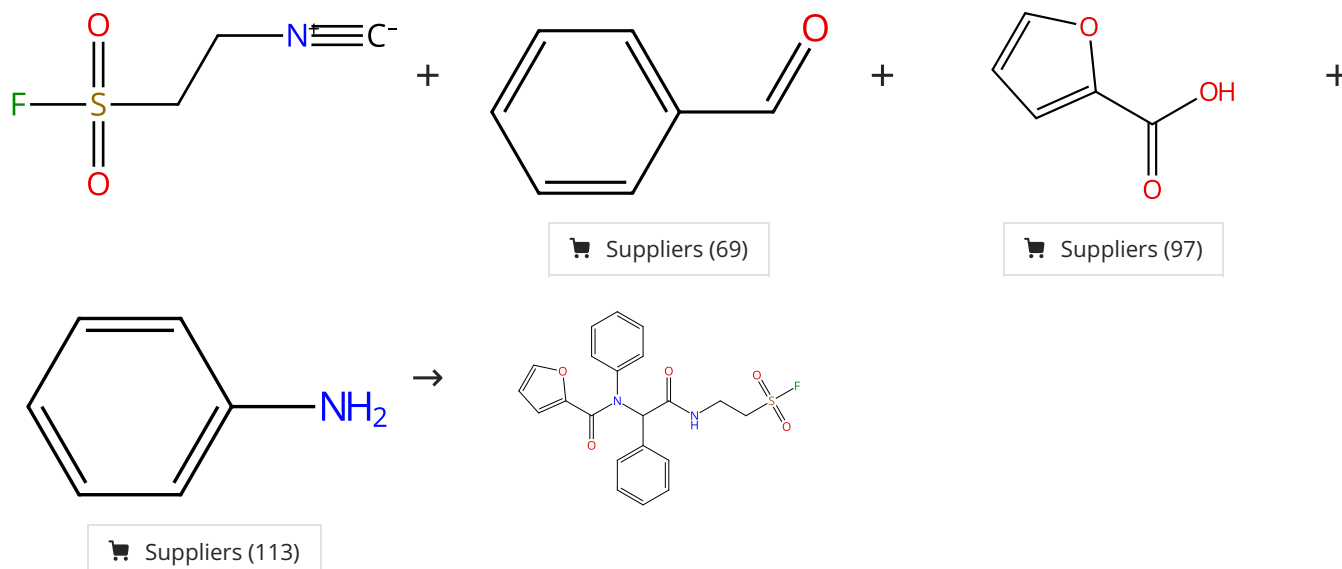
By: Xu, Shuheng; et al

Experimental Protocols

Organic Letters (2021), 23(13), 5197-5202.

Scheme 5 (1 Reaction)

Steps: 1 Yield: 99%



31-367-CAS-23902249

Steps: 1 Yield: 99%

SuFExable isocyanides for Ugi reaction: Synthesis of sulfonyl fluoro peptides

1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 3 h, rt

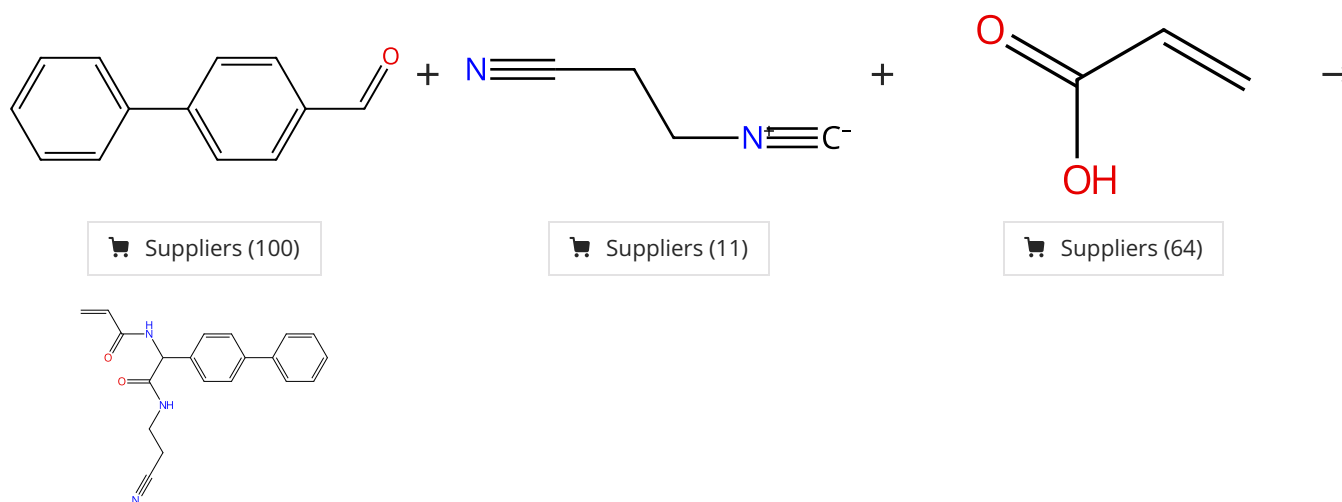
Experimental Protocols

By: Xu, Shuheng; et al

Organic Letters (2021), 23(13), 5197-5202.

Scheme 6 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-34665224

Steps: 1 Yield: 99%

Combining High-Throughput Synthesis and High-Throughput Protein Crystallography for Accelerated Hit Identification

1.1 Reagents: Ammonium hydroxide

Solvents: 2,2,2-Trifluoroethanol, Water; 15 min, rt

1.2 overnight, rt

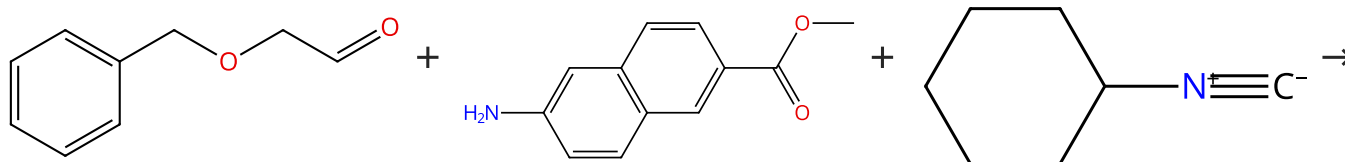
Experimental Protocols

By: Sutanto, Fandi; et al

Angewandte Chemie, International Edition (2021), 60(33), 18231-18239.

Scheme 7 (1 Reaction)

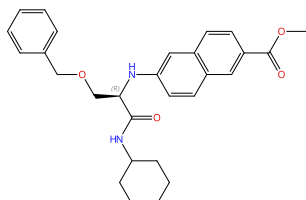
Steps: 1 Yield: 99%



Suppliers (91)

Suppliers (66)

Suppliers (72)



Absolute stereochemistry shown

31-389-CAS-21608916

Steps: 1 Yield: 99%

Enantioselective three-component Ugi reaction catalyzed by chiral phosphoric acid

By: Zhang, Jian; et al

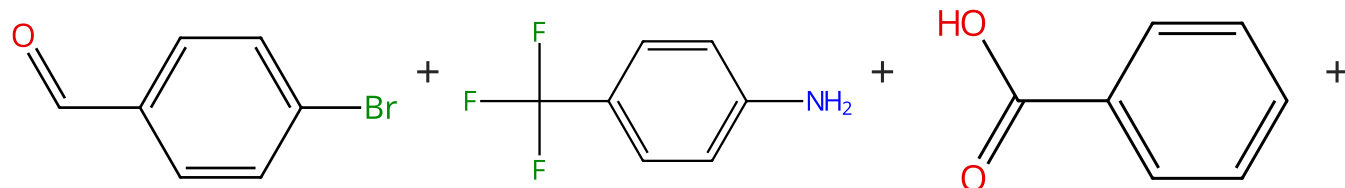
Science China: Chemistry (2020), 63(1), 47-54.

1.1 **Catalysts:** Diindeno[7,1-*de*:1',7'-*fg*][1,3,2]dioxaphosphocin, 10, 11,12,13-tetrahydro-5-hydroxy-3,7-bis(2,4,6-tricyclohexylphenyl)-, 5-oxide, (11a5)-
Solvents: Dichloromethane; -30 °C; 36 h, -30 °C

Experimental Protocols

Scheme 8 (1 Reaction)

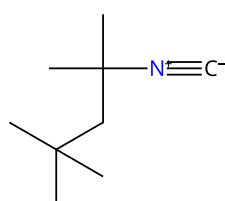
Steps: 1 Yield: 99%



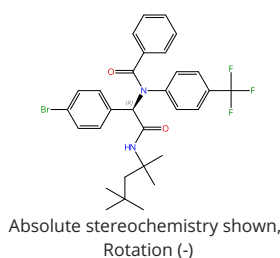
Suppliers (91)

Suppliers (86)

Suppliers (175)



Suppliers (57)



Absolute stereochemistry shown, Rotation (-)

31-614-CAS-34981578

Steps: 1 Yield: 99%

Enantioselective Ugi and Ugi-azide reactions catalyzed by anionic stereogenic-at-cobalt(III) complexes

By: Sun, Bing-Bing; et al

Nature Communications (2022), 13(1), 7065.

1.1 **Catalysts:** Cobaltate(1-), bis[*N*-[[3,5-bis(1,1-dimethylethyl)-2-(hydroxy- κ O)phenyl]methylene]-3-methyl-L-valinato(2-)- κ N, κ O]-, sodium (1:1), (*OC*-6-22'-*C*)-
Solvents: Toluene; 30 min, rt

1.2 30 min, rt $\rightarrow$ -40 °C

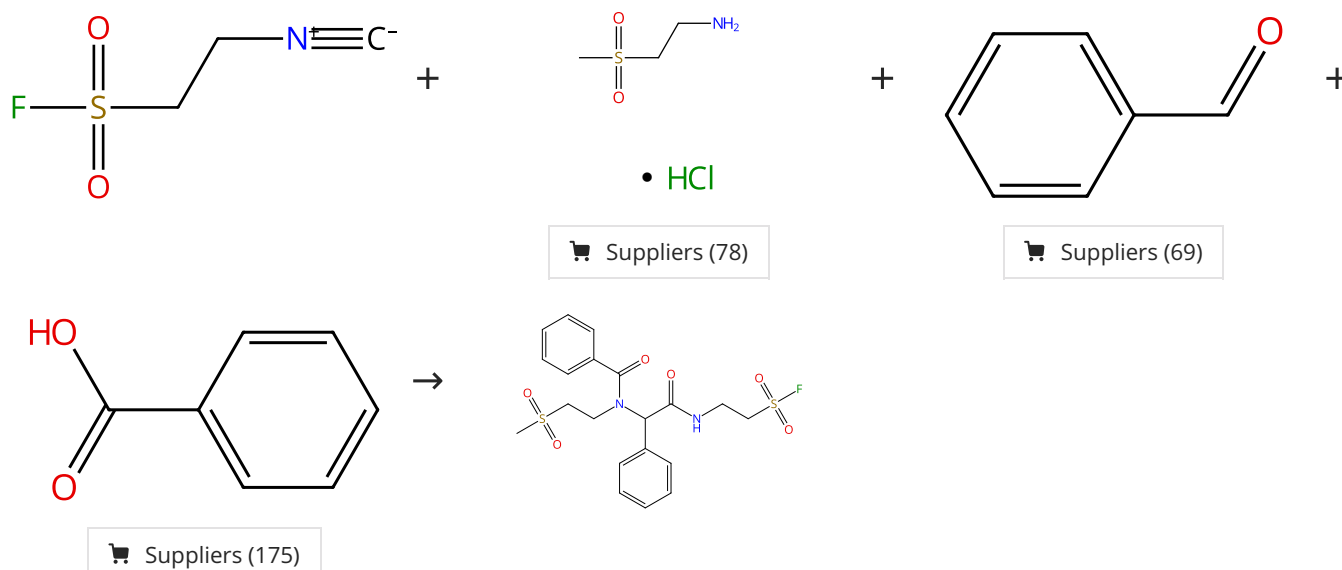
1.3 24 h, -40 °C

1.4 **Reagents:** Triethylamine; -40 °C

Experimental Protocols

Scheme 9 (1 Reaction)

Steps: 1 Yield: 99%



31-367-CAS-23905471

Steps: 1 Yield: 99%

SuFExable isocyanides for Ugi reaction: Synthesis of sulfonyl fluoro peptides

1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 3 h, rt

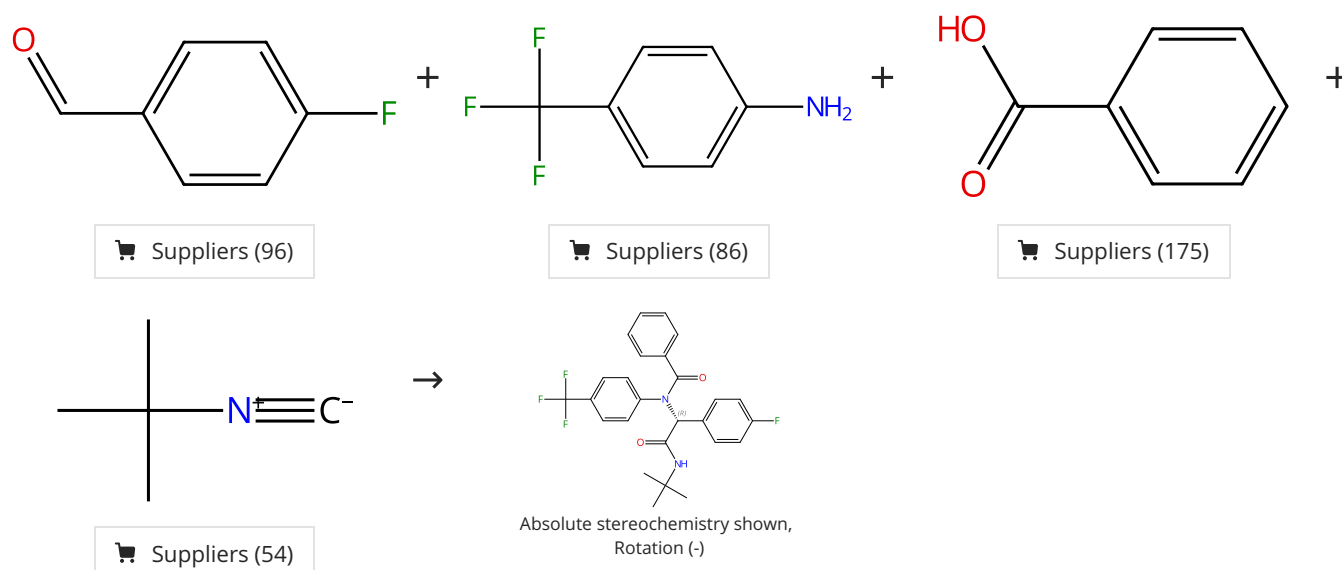
Experimental Protocols

By: Xu, Shuheng; et al

Organic Letters (2021), 23(13), 5197-5202.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-34981558

Steps: 1 Yield: 99%

Enantioselective Ugi and Ugi-azide reactions catalyzed by anionic stereogenic-at-cobalt(III) complexes

1.1 Catalysts: Cobaltate(1-), bis[*N*-[[3,5-bis(1,1-dimethylethyl)-2-(hydroxy-κ*O*)phenyl]methylene]-3-methyl-L-valinato(2-)-κ*N*, κ*O*]-, sodium (1:1), (*OC*-6-22'-*O*)-

Solvents: Toluene; 30 min, rt

1.2 30 min, rt → -40 °C

1.3 24 h, -40 °C

1.4 Reagents: Triethylamine; -40 °C

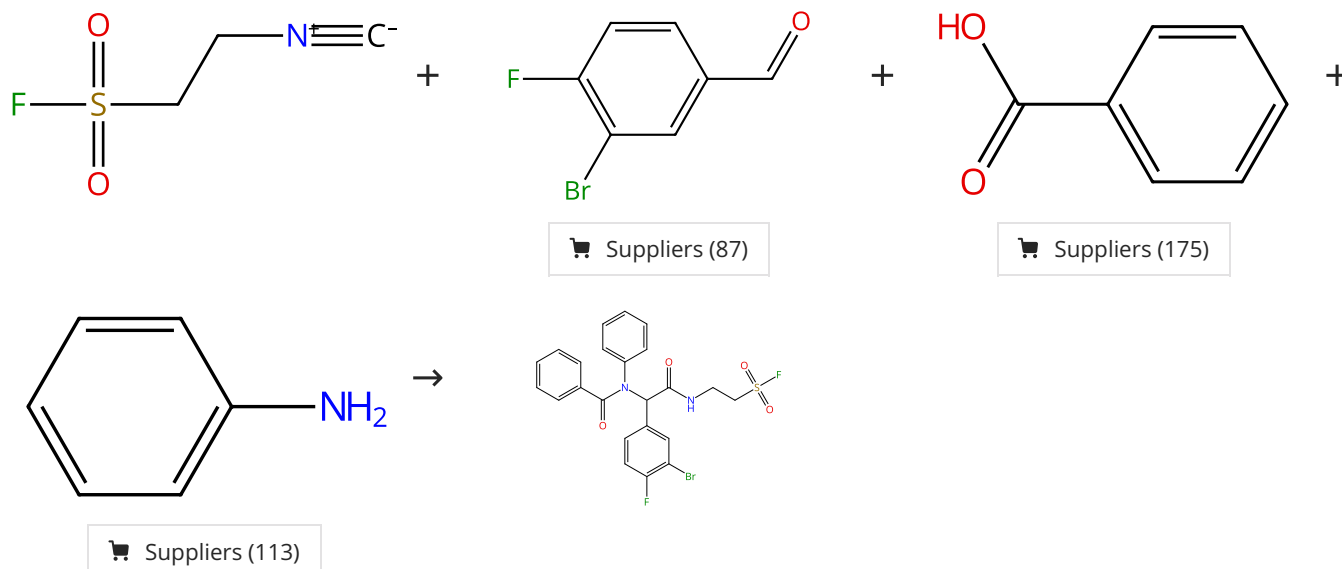
Experimental Protocols

By: Sun, Bing-Bing; et al

Nature Communications (2022), 13(1), 7065.

Scheme 11 (1 Reaction)

Steps: 1 Yield: 99%



31-367-CAS-23902577

Steps: 1 Yield: 99%

SuFExable isocyanides for Ugi reaction: Synthesis of sulfonyl fluoro peptides

1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 3 h, rt

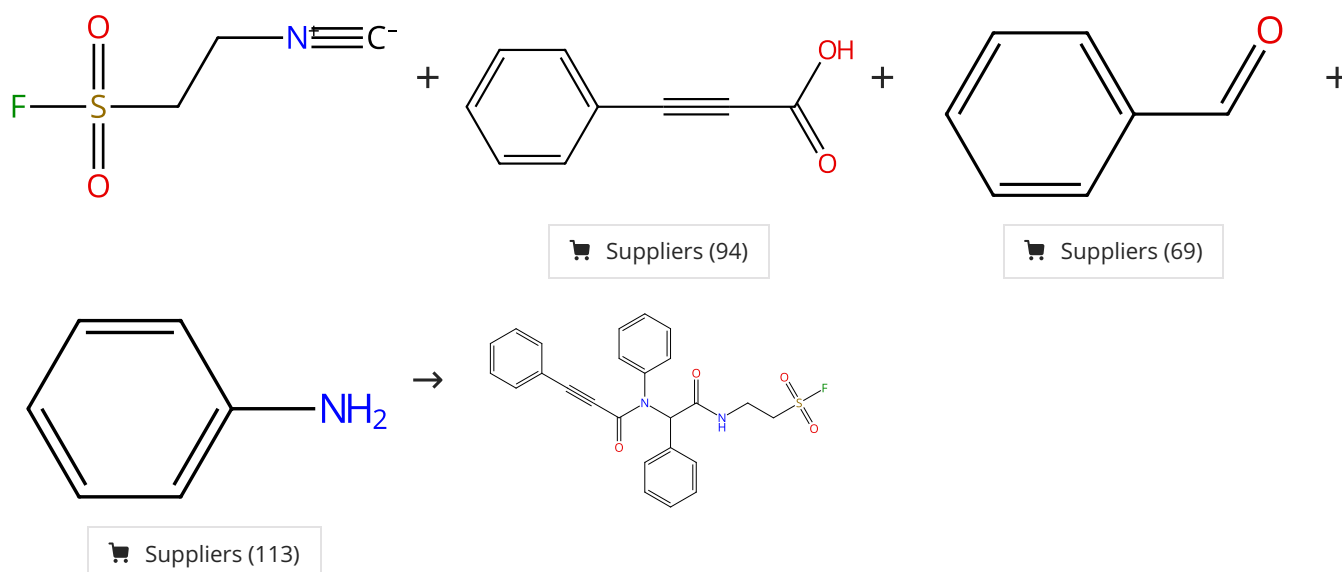
By: Xu, Shuheng; et al

Experimental Protocols

Organic Letters (2021), 23(13), 5197-5202.

Scheme 12 (1 Reaction)

Steps: 1 Yield: 99%



31-367-CAS-23904845

Steps: 1 Yield: 99%

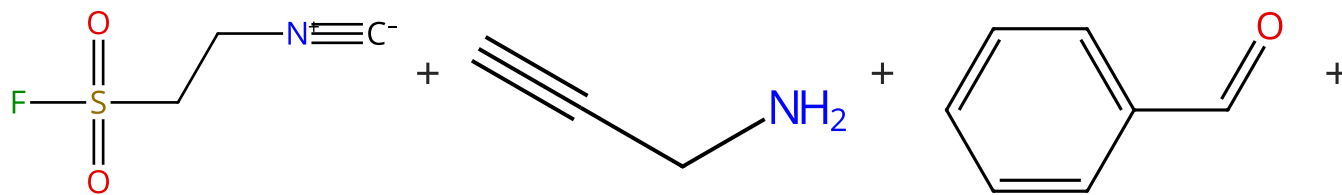
SuFExable isocyanides for Ugi reaction: Synthesis of sulfonyl fluoro peptides

1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 3 h, rt

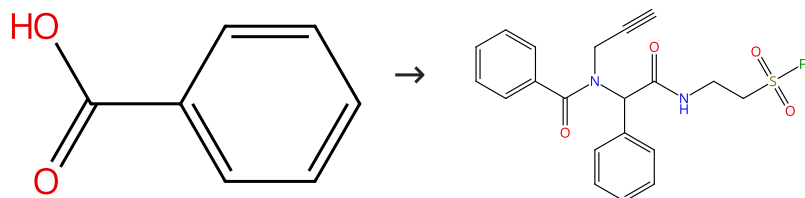
By: Xu, Shuheng; et al

Experimental Protocols

Organic Letters (2021), 23(13), 5197-5202.

Steps: **1** Yield: **99%**

 Suppliers (69)



 Suppliers (175)

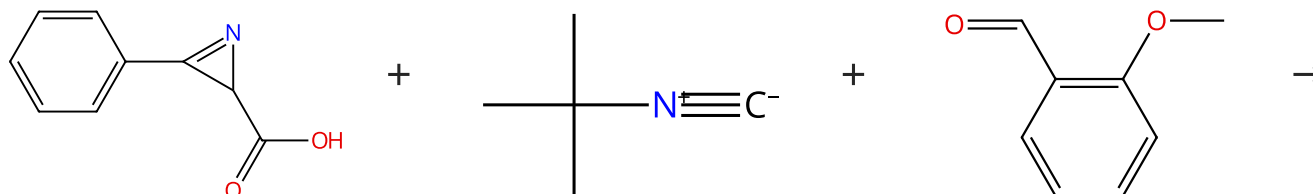
Steps: **1** Yield: **99%**

SuFExable isocyanides for Ugi reaction: Synthesis of sulfonyl fluoro peptides

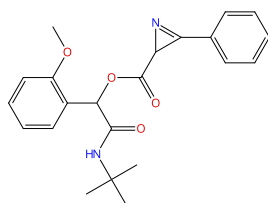
By: Xu, Shuheng; et al

Organic Letters (2021), 23(13), 5197-5202.

Steps: **1** Yield: **99%**



 Suppliers (96)

Steps: **1** Yield: **99%**

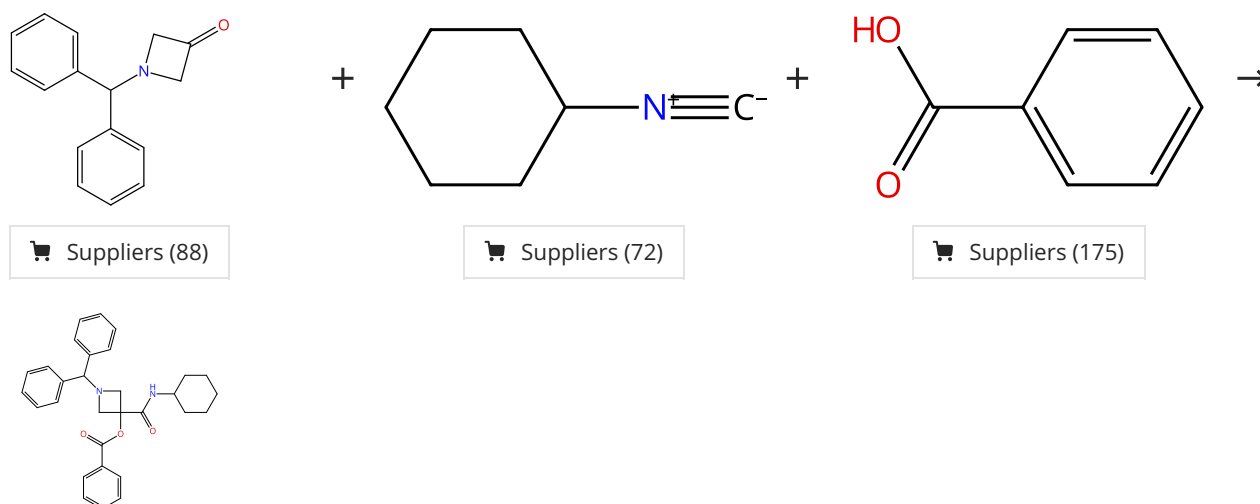
Azirine-containing dipeptides and depsipeptides: Synthesis, transformations and antibacterial activity

By: Rostovskii, Nikolai V.; et al

Organic & Biomolecular Chemistry (2020), 18(46), 9448-9460.

Scheme 15 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-37350913

Steps: 1 Yield: 99%

Strain and Complexity, Passerini and Ugi Reactions of Four-Membered Heterocycles and Further Elaboration of TOSMIC Product

By: Sztano, Gabor; et al

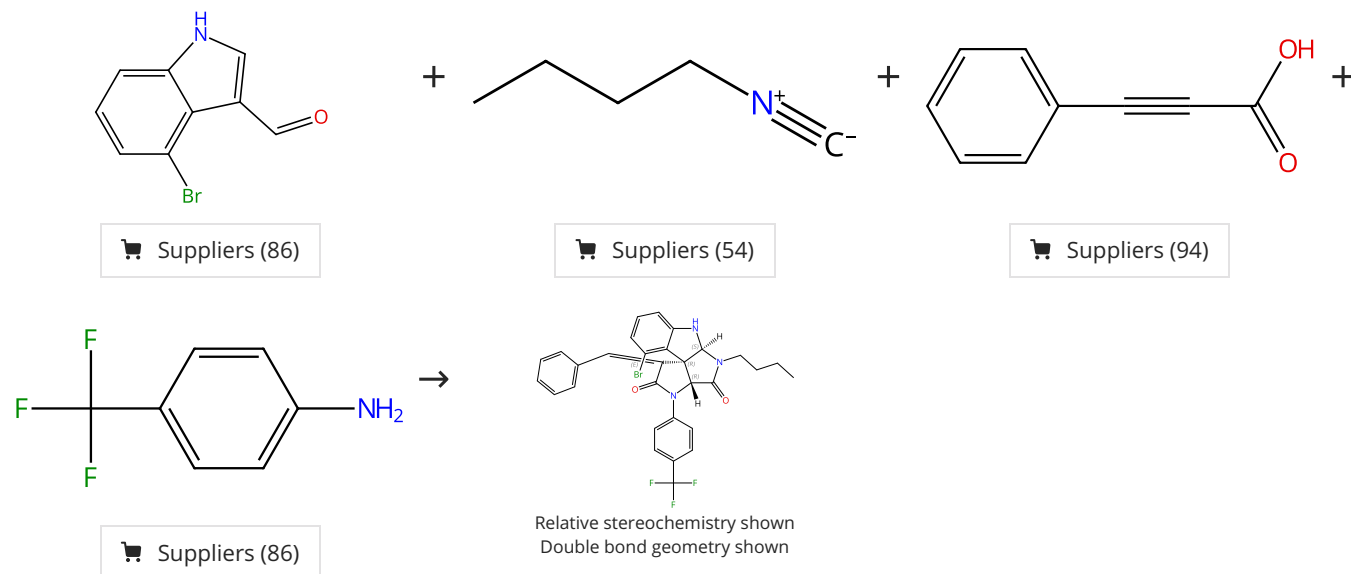
ChemistryOpen (2023), 12(8), e202200083.

1.1 Solvents: Toluene; 2 d, 40 °C

Experimental Protocols

Scheme 16 (1 Reaction)

Steps: 1 Yield: 99%



31-367-CAS-21006789

Steps: 1 Yield: 99%

Silver(I) Triflate-Catalyzed Protocol for the Post-Ugi Synthesis of Spiroindolines

By: Zaman, Manzoor; et al

Advanced Synthesis & Catalysis (2020), 362(1), 261-268.

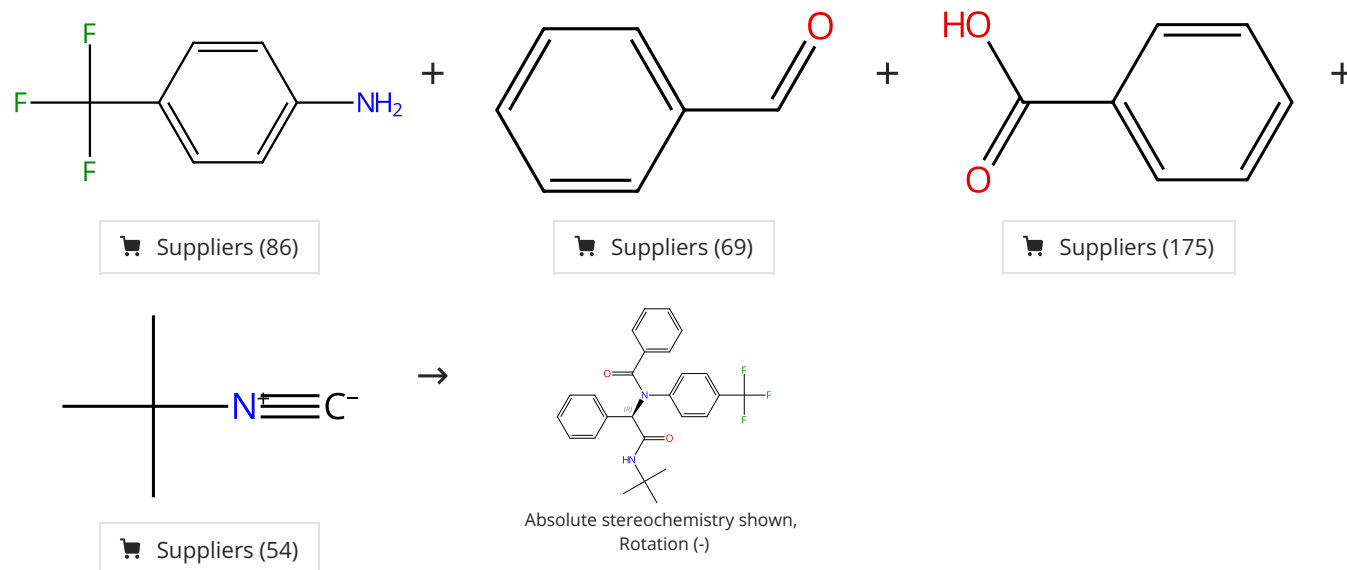
1.1 Solvents: Methanol; 24 h, rt

1.2 Reagents: Trifluoroacetic acid
Catalysts: Silver triflate; 24 h, rt

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-34981573

Steps: 1 Yield: 99%

Enantioselective Ugi and Ugi-azide reactions catalyzed by anionic stereogenic-at-cobalt(III) complexes

By: Sun, Bing-Bing; et al

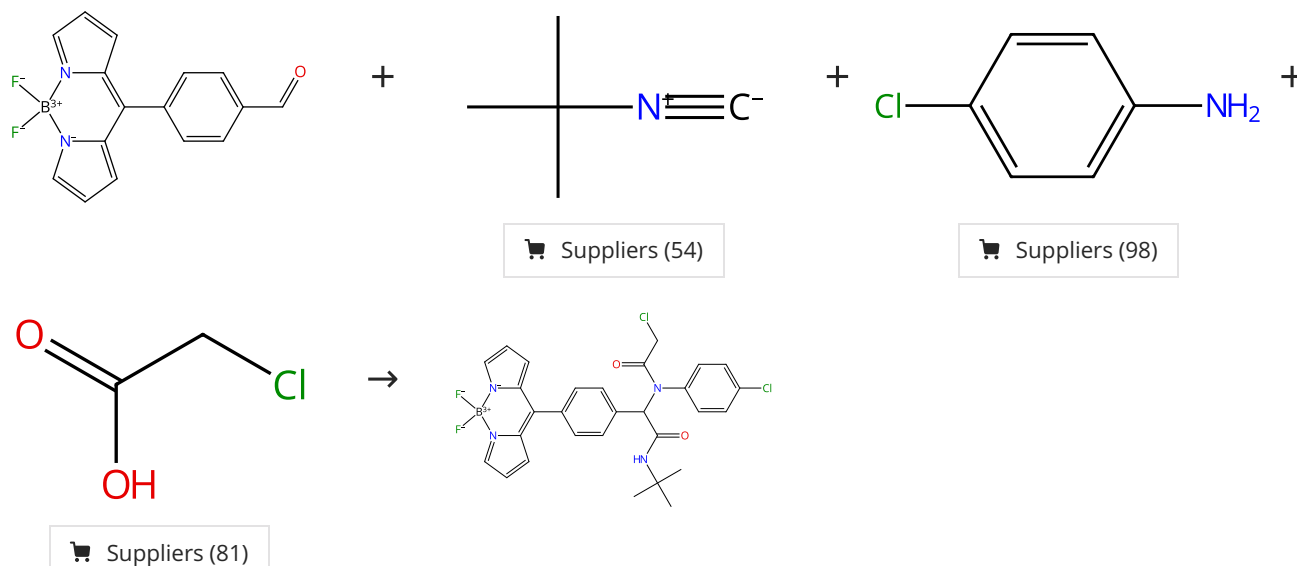
Nature Communications (2022), 13(1), 7065.

- 1.1 **Catalysts:** Cobaltate(1-), bis[*N*-[[3,5-bis(1,1-dimethylethyl)-2-(hydroxy- κ O)phenyl]methylene]-3-methyl-L-valinato(2-)- κ N, κ O]-, sodium (1:1), (*OC*-6-22'-*C*)-
Solvents: Toluene; 30 min, rt
 1.2 30 min, rt $\rightarrow$ -40 °C
 1.3 24 h, -40 °C
 1.4 **Reagents:** Triethylamine; -40 °C

Experimental Protocols

Scheme 18 (1 Reaction)

Steps: 1 Yield: 99%



31-367-CAS-23060916

Steps: 1 Yield: 99%

Mechanochemistry as a Sustainable Method for the Preparation of Fluorescent Ugi BODIPY Adducts

By: Perez-Venegas, Mario; et al

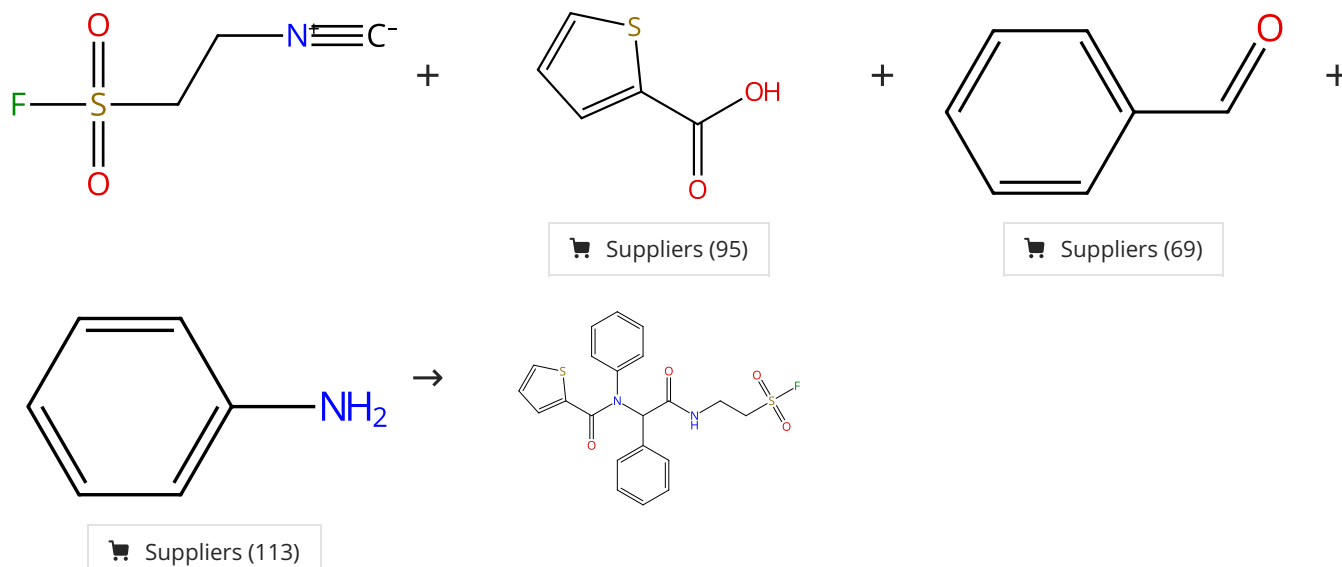
European Journal of Organic Chemistry (2021), 2021(2), 253-265.

- 1.1 **Catalysts:** Indium trichloride
Solvents: Methanol; 90 min, rt

Experimental Protocols

Scheme 19 (1 Reaction)

Steps: 1 Yield: 99%



31-367-CAS-23904166

Steps: 1 Yield: 99%

SuFExable isocyanides for Ugi reaction: Synthesis of sulfonyl fluoro peptides

1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 3 h, rt

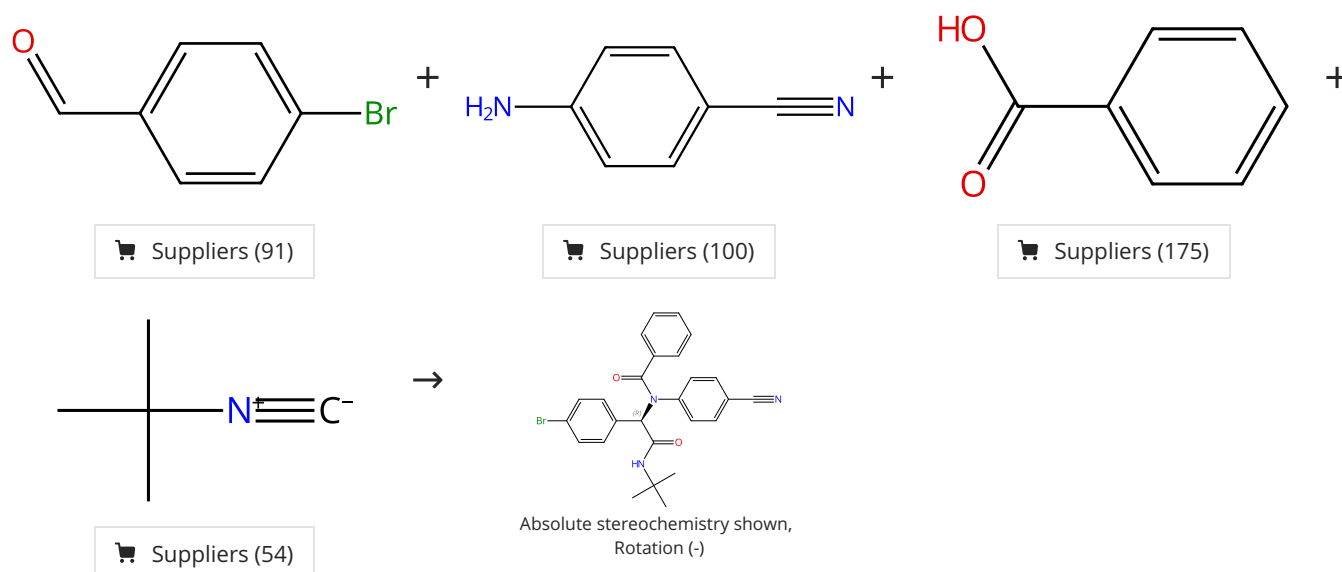
Experimental Protocols

By: Xu, Shuheng; et al

Organic Letters (2021), 23(13), 5197-5202.

Scheme 20 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-34981531

Steps: 1 Yield: 99%

Enantioselective Ugi and Ugi-azide reactions catalyzed by anionic stereogenic-at-cobalt(III) complexes

1.1 Catalysts: Cobaltate(1-), bis[*N*-[[3,5-bis(1,1-dimethylethyl)-2-(hydroxy-κ*O*)phenyl]methylene]-3-methyl-L-valinato(2-)-κ*N*, κ*O*]-, sodium (1:1), (*OC*-6-22'-*O*)-

Solvents: Toluene; 30 min, rt

1.2 30 min, rt → -40 °C

1.3 24 h, -40 °C

1.4 Reagents: Triethylamine; -40 °C

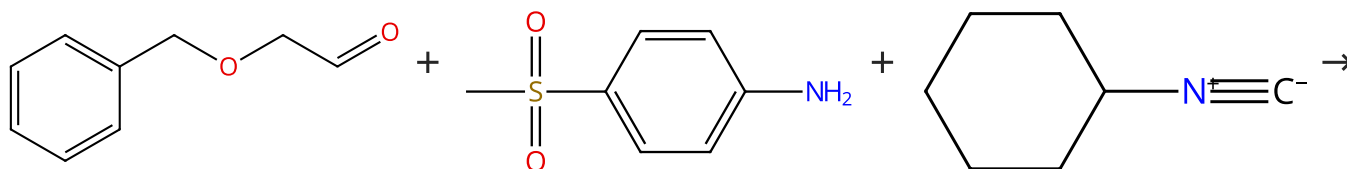
Experimental Protocols

By: Sun, Bing-Bing; et al

Nature Communications (2022), 13(1), 7065.

Scheme 21 (1 Reaction)

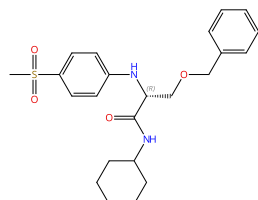
Steps: 1 Yield: 99%



Suppliers (91)

Suppliers (79)

Suppliers (72)



Absolute stereochemistry shown

31-389-CAS-21608908

Steps: 1 Yield: 99%

Enantioselective three-component Ugi reaction catalyzed by chiral phosphoric acid

By: Zhang, Jian; et al

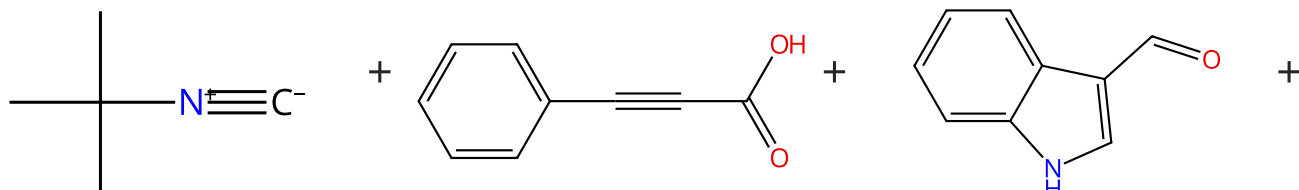
Science China: Chemistry (2020), 63(1), 47-54.

1.1 **Catalysts:** Diindenod[7,1-*de*:1',7'-*fg*][1,3,2]dioxaphosphocin, 10, 11,12,13-tetrahydro-5-hydroxy-3,7-bis(2,4,6-tricyclohexylphenyl)-, 5-oxide, (11a5)-
Solvents: Dichloromethane; -30 °C; 36 h, -30 °C

Experimental Protocols

Scheme 22 (1 Reaction)

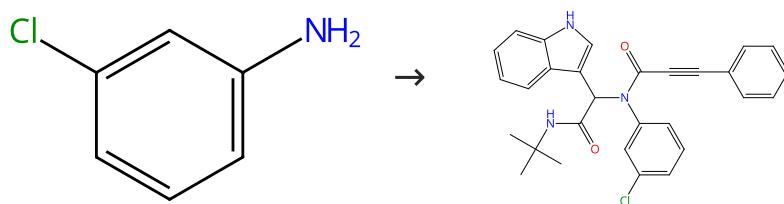
Steps: 1 Yield: 99%



Suppliers (54)

Suppliers (94)

Suppliers (116)



Suppliers (75)

31-367-CAS-21006756

Steps: 1 Yield: 99%

Silver(I) Triflate-Catalyzed Protocol for the Post-Ugi Synthesis of Spiroindolines

By: Zaman, Manzoor; et al

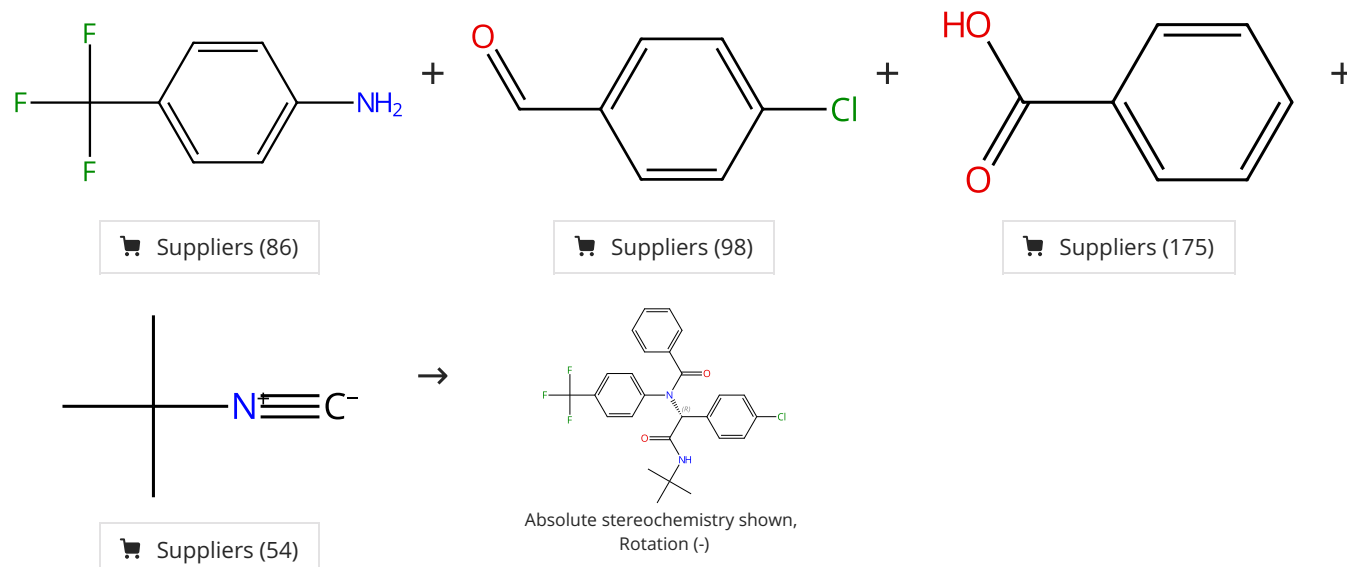
Advanced Synthesis & Catalysis (2020), 362(1), 261-268.

1.1 **Solvents:** Methanol; 24 h, rt

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-34981565

Steps: 1 Yield: 99%

Enantioselective Ugi and Ugi-azide reactions catalyzed by anionic stereogenic-at-cobalt(III) complexes

By: Sun, Bing-Bing; et al

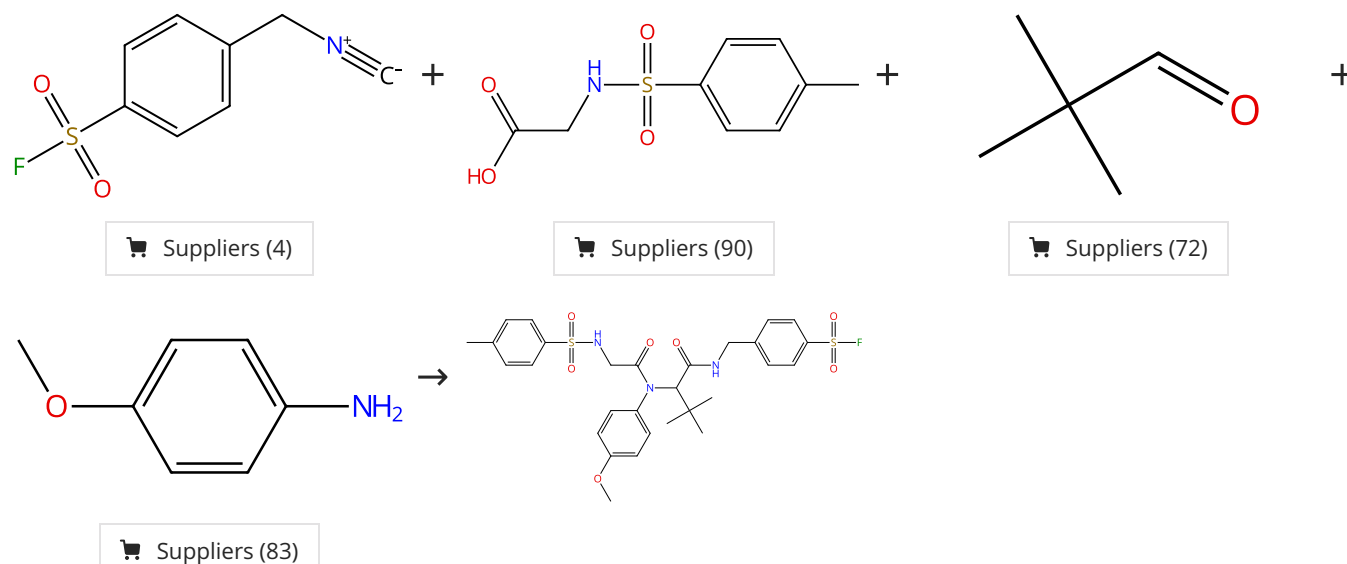
Nature Communications (2022), 13(1), 7065.

- 1.1 **Catalysts:** Cobaltate(1-), bis[*N*-[[3,5-bis(1,1-dimethylethyl)-2-(hydroxy- κO)phenyl]methylene]-3-methyl-L-valinato(2-)- $\kappa N, \kappa O$]-, sodium (1:1), (*OC*-6-22'-*C*)-
Solvents: Toluene; 30 min, rt
 1.2 30 min, rt $\rightarrow$ -40 °C
 1.3 24 h, -40 °C
 1.4 **Reagents:** Triethylamine; -40 °C

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 99%



31-367-CAS-23904662

Steps: 1 Yield: 99%

SuFExable isocyanides for Ugi reaction: Synthesis of sulfonyl fluoro peptides

By: Xu, Shuheng; et al

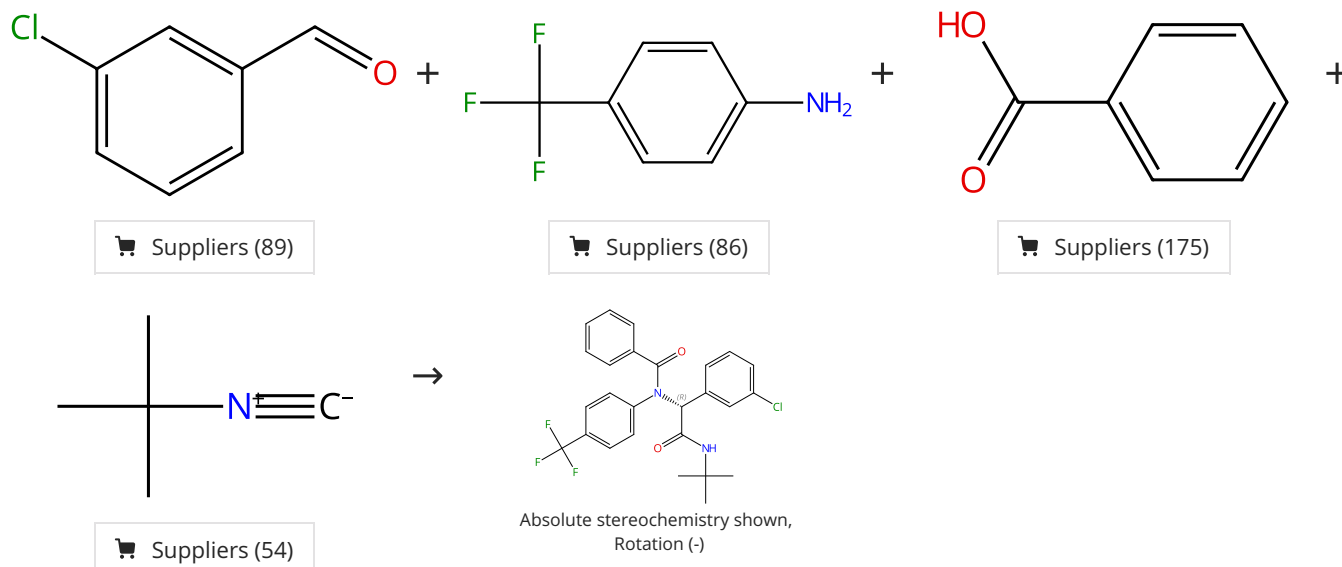
Organic Letters (2021), 23(13), 5197-5202.

- 1.1 **Solvents:** 1,1,1,3,3,3-Hexafluoro-2-propanol; 3 h, rt

Experimental Protocols

Scheme 25 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-34981559

Steps: 1 Yield: 99%

Enantioselective Ugi and Ugi-azide reactions catalyzed by anionic stereogenic-at-cobalt(III) complexes

By: Sun, Bing-Bing; et al

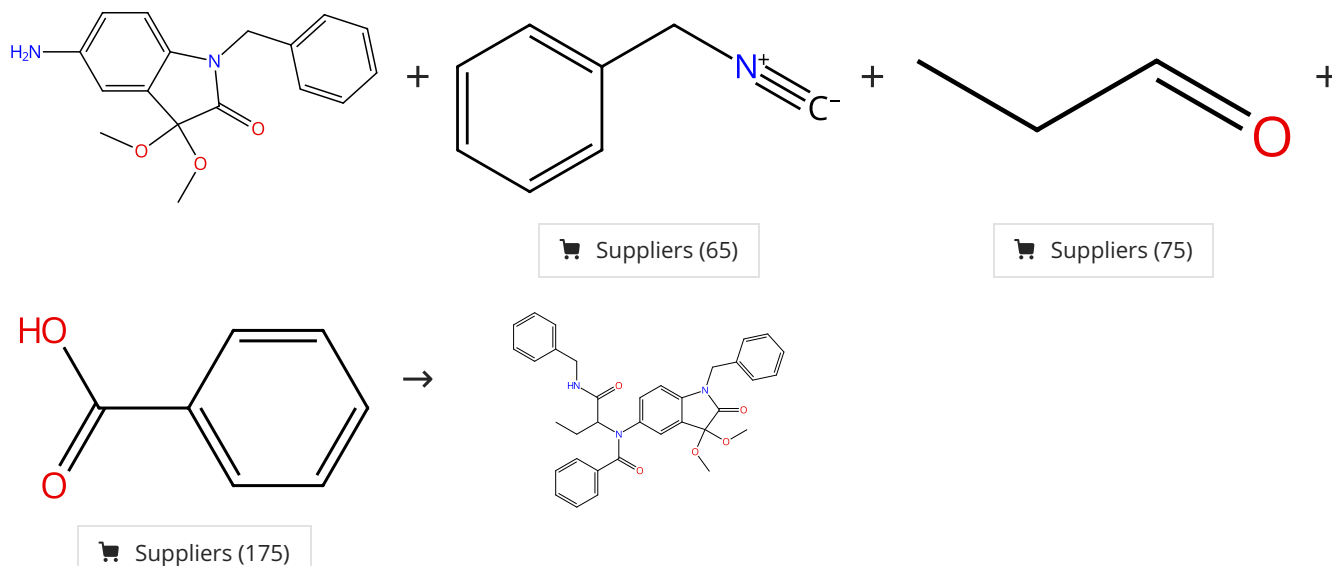
Nature Communications (2022), 13(1), 7065.

- 1.1 **Catalysts:** Cobaltate(1-), bis[*N*-[[3,5-bis(1,1-dimethylethyl)-2-(hydroxy- κO)phenyl]methylene]-3-methyl-L-valinato(2-)- κN , κO]-, sodium (1:1), (*OC*-6-22'-*C*)-
Solvents: Toluene; 30 min, rt
- 1.2 30 min, rt $\rightarrow$ -40 °C
- 1.3 24 h, -40 °C
- 1.4 **Reagents:** Triethylamine; -40 °C

Experimental Protocols

Scheme 26 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-35204285

Steps: 1 Yield: 99%

Easy access to Ugi-derived isatin-peptoids and their potential as small-molecule anticancer agents

By: Marques, Carolina S.; et al

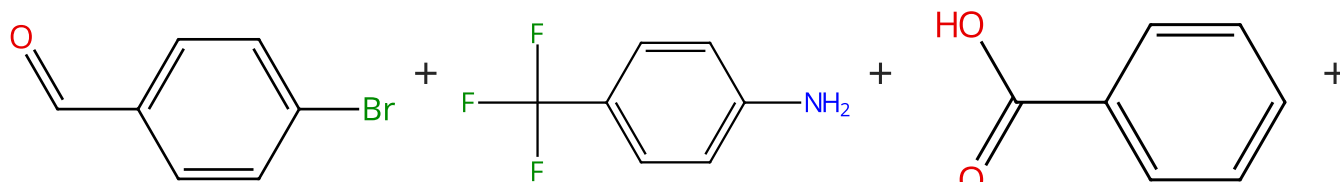
New Journal of Chemistry (2023), 47(2), 743-750.

- 1.1 **Catalysts:** Zinc fluoride
Solvents: Methanol; 2 h, rt

Experimental Protocols

Scheme 27 (1 Reaction)

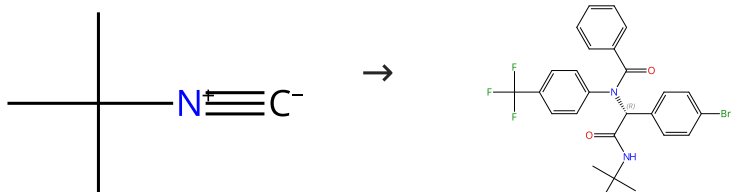
Steps: 1 Yield: 99%



Suppliers (91)

Suppliers (86)

Suppliers (175)



Suppliers (54)

Absolute stereochemistry shown,
Rotation (-)

31-614-CAS-34981536

Steps: 1 Yield: 99%

Enantioselective Ugi and Ugi-azide reactions catalyzed by anionic stereogenic-at-cobalt(III) complexes

By: Sun, Bing-Bing; et al

Nature Communications (2022), 13(1), 7065.

1.1 **Catalysts:** Cobaltate(1-), bis[*N*-[[3,5-bis(1,1-dimethylethyl)-2-(hydroxy-κ*O*)phenyl]methylene]-3-methyl-L-valinato(2-)-κ*N*, κ*O*]-, sodium (1:1), (*OC*-6-22'-*C*)-
Solvents: Toluene; 30 min, rt

1.2 30 min, rt → -40 °C

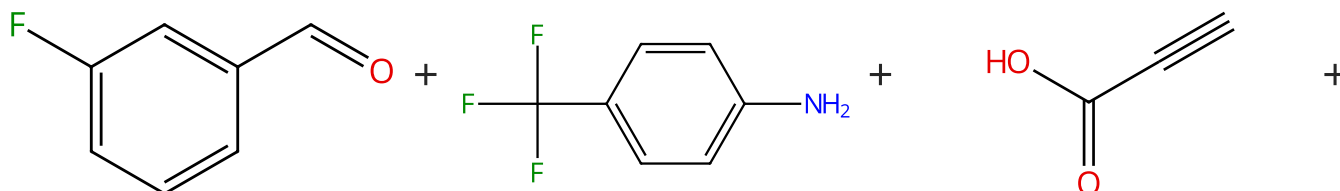
1.3 24 h, -40 °C

1.4 **Reagents:** Triethylamine; -40 °C

Experimental Protocols

Scheme 28 (1 Reaction)

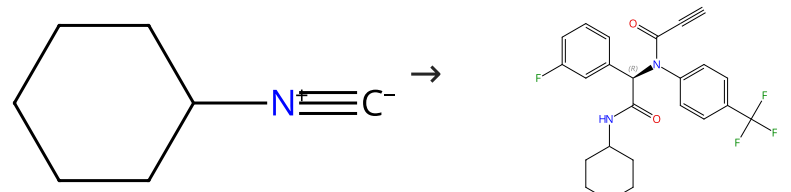
Steps: 1 Yield: 99%



Suppliers (82)

Suppliers (86)

Suppliers (61)



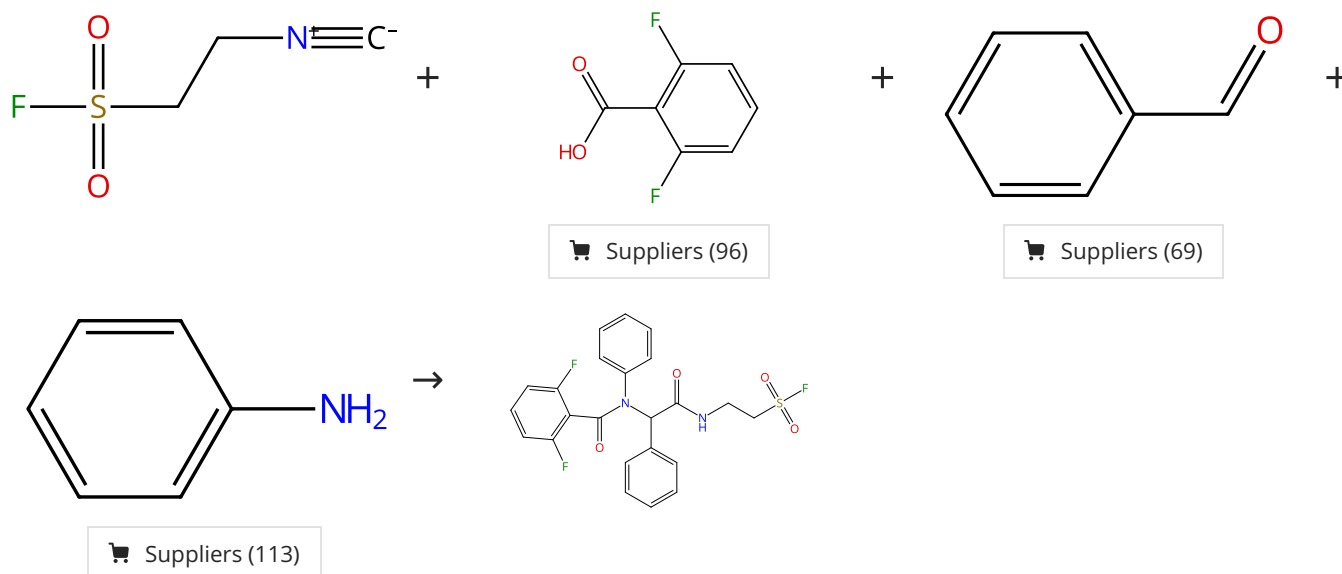
Suppliers (72)

Absolute stereochemistry shown,
Rotation (-)

31-614-CAS-38204870 Steps: 1 Yield: 99% 1.1 Catalysts: Cobaltate(1-), bis[<i>N</i>-[[3,5-bis(1,1-dimethylethyl)-2-(hydroxy-κO)phenyl]methylene]-3-methyl-L-valinato(2-)-κN, κO]-, sodium (1:1), (<i>OC</i>-6-22'-<i>O</i>)- Solvents: Toluene; 30 min, rt 1.2 rt $\rightarrow$ -50 °C; 30 min, -50 °C 1.3 12 h, -50 °C 1.4 Reagents: Sodium bicarbonate Solvents: Water Experimental Protocols	Enantioselective Access to Chiral 2,5-Diketopiperazines via Stereogenic-at-Cobalt(III)-Catalyzed Ugi-4CRs/Cyclization Sequences By: Fang, Wei; et al Journal of Organic Chemistry (2023), 88(22), 16024-16037.
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Scheme 29 (1 Reaction)

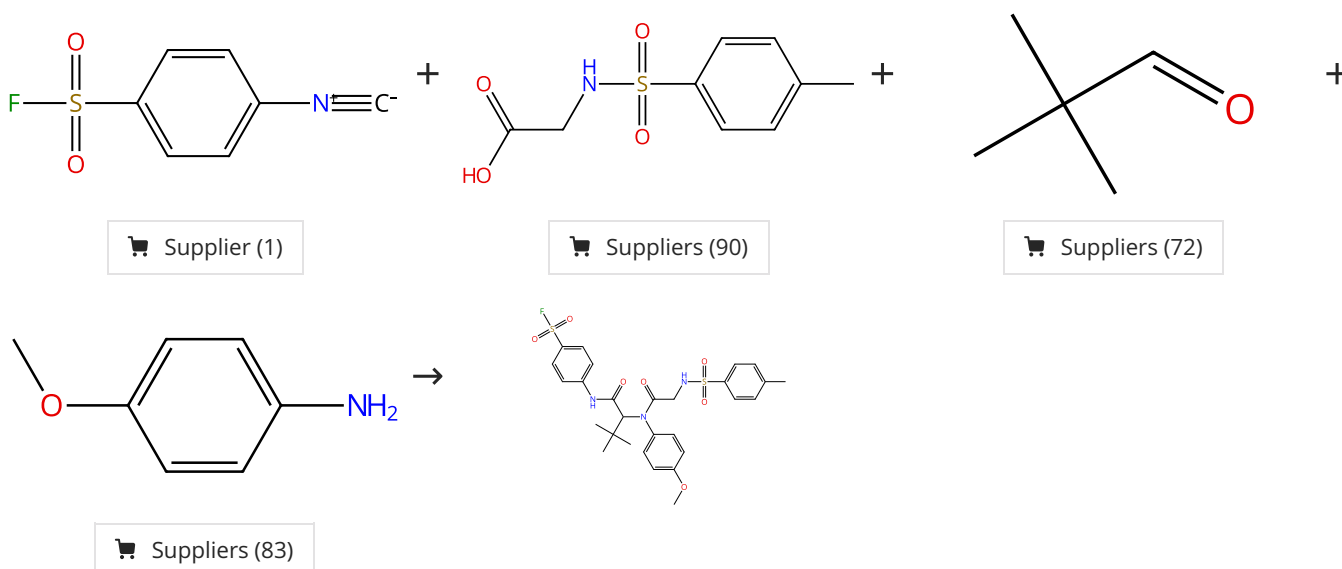
Steps: 1 Yield: 99%



31-367-CAS-23904998 Steps: 1 Yield: 99% 1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 3 h, rt Experimental Protocols	SuFExable isocyanides for Ugi reaction: Synthesis of sulfonyl fluoro peptides By: Xu, Shuheng; et al Organic Letters (2021), 23(13), 5197-5202.
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Scheme 30 (1 Reaction)

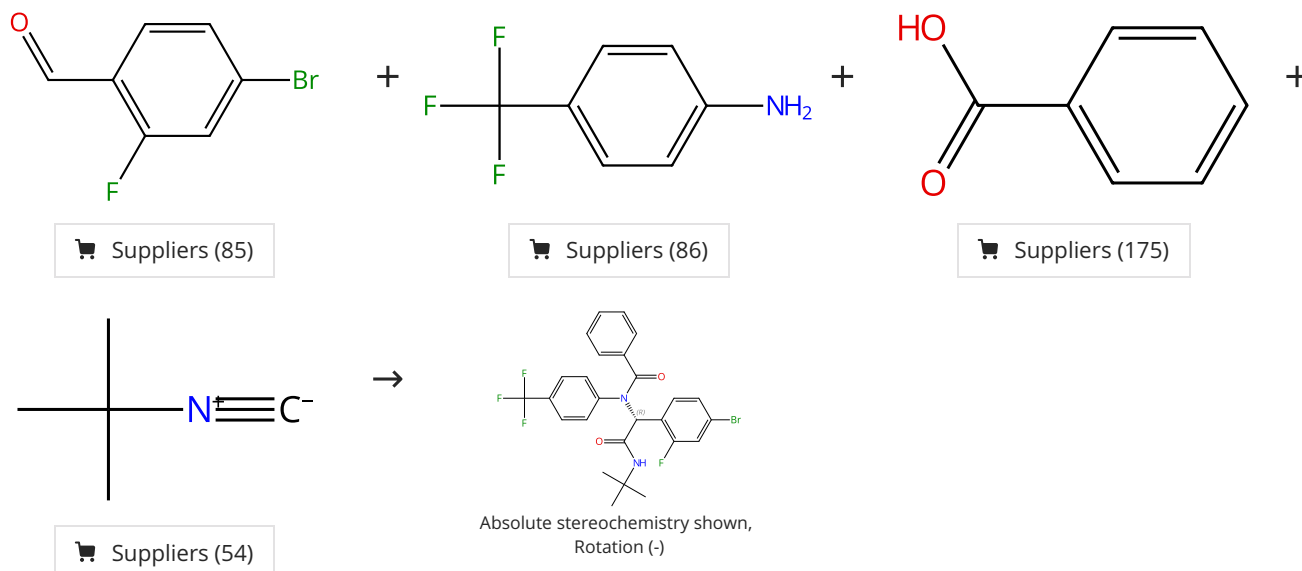
Steps: 1 Yield: 99%



31-367-CAS-23905285	Steps: 1 Yield: 99%	SuFExable isocyanides for Ugi reaction: Synthesis of sulfonyl fluoro peptides
1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 3 h, rt		By: Xu, Shuheng; et al
Experimental Protocols		Organic Letters (2021), 23(13), 5197-5202.

Scheme 31 (1 Reaction)

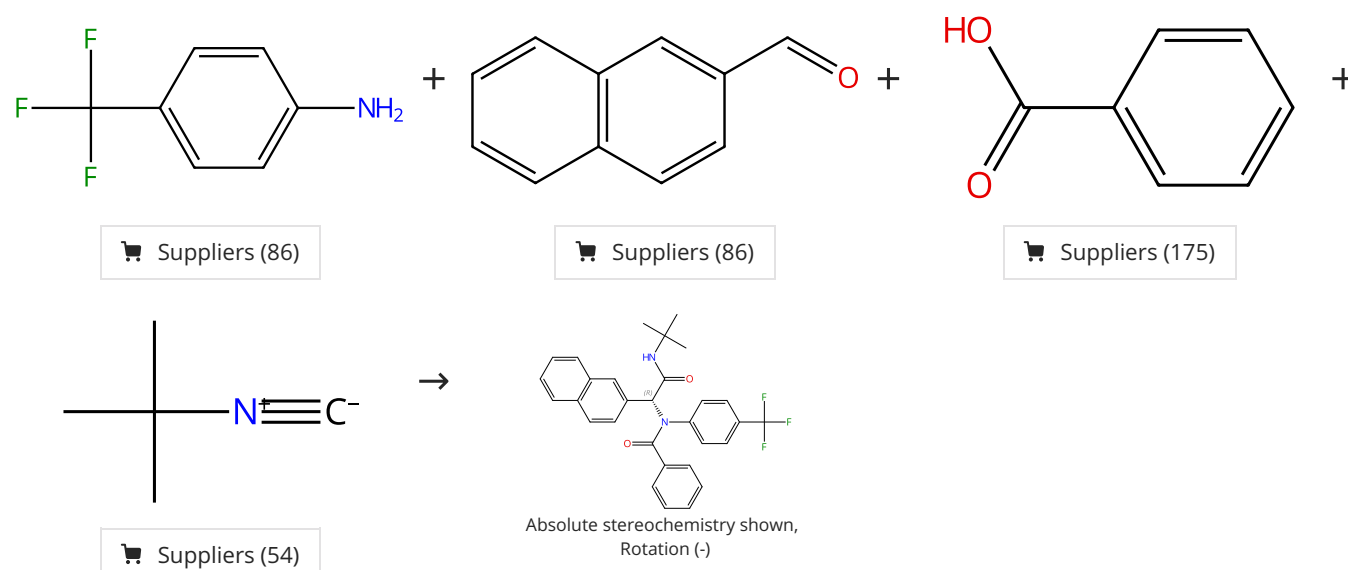
Steps: 1 Yield: 99%



31-614-CAS-34981562	Steps: 1 Yield: 99%	Enantioselective Ugi and Ugi-azide reactions catalyzed by anionic stereogenic-at-cobalt(III) complexes
1.1 Catalysts: Cobaltate(1-), bis[<i>N</i> -[[3,5-bis(1,1-dimethylethyl)-2-(hydroxy- κ O)phenyl]methylene]-3-methyl-L-valinato(2-)- κ N, κ O]-, sodium (1:1), (<i>OC</i> -6-22'- <i>O</i>)-		By: Sun, Bing-Bing; et al
Solvents: Toluene; 30 min, rt		Nature Communications (2022), 13(1), 7065.
1.2 30 min, rt $\rightarrow$ -40 °C		
1.3 24 h, -40 °C		
1.4 Reagents: Triethylamine; -40 °C		
Experimental Protocols		

Scheme 32 (1 Reaction)

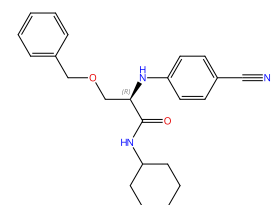
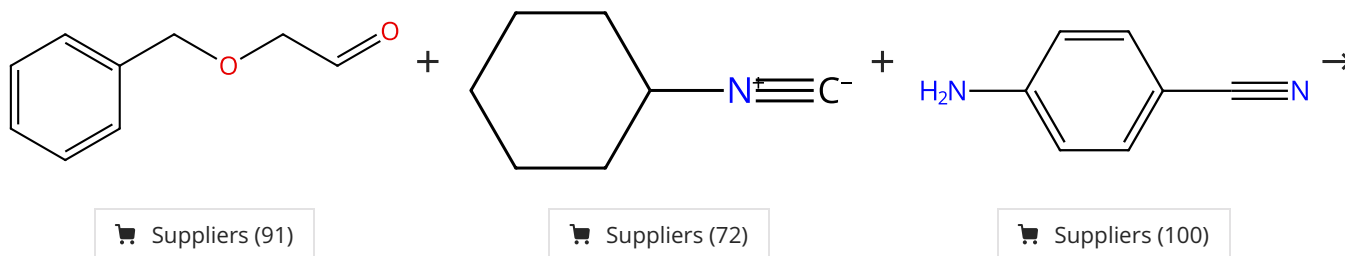
Steps: 1 Yield: 99%



31-614-CAS-34981571 Steps: 1 Yield: 99% 1.1 Catalysts: Cobaltate(1-), bis[<i>N</i>-[[3,5-bis(1,1-dimethylethyl)-2-(hydroxy-κO)phenyl]methylene]-3-methyl-L-valinato(2-)-κN, κO]-, sodium (1:1), (<i>OC</i>-6-22'-<i>O</i>)- Solvents: Toluene; 30 min, rt 1.2 30 min, rt $\rightarrow$ -40 °C 1.3 24 h, -40 °C 1.4 Reagents: Triethylamine; -40 °C Experimental Protocols	Enantioselective Ugi and Ugi-azide reactions catalyzed by anionic stereogenic-at-cobalt(III) complexes By: Sun, Bing-Bing; et al Nature Communications (2022), 13(1), 7065.
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Scheme 33 (1 Reaction)

Steps: 1 Yield: 99%

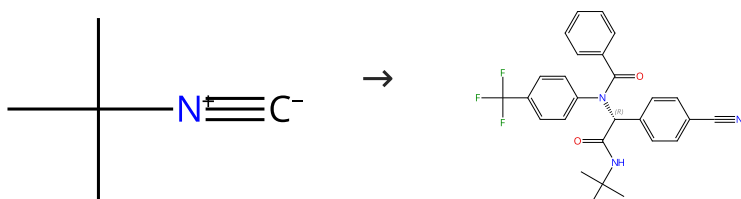
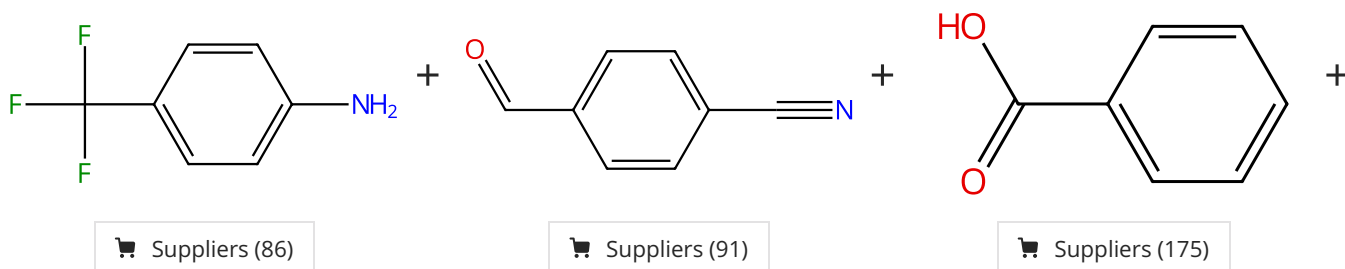


Absolute stereochemistry shown

31-389-CAS-21608909 Steps: 1 Yield: 99% 1.1 Catalysts: Diindeno[7,1-<i>de</i>:1',7'-<i>fg</i>][1,3,2]dioxaphosphocin, 10, 11,12,13-tetrahydro-5-hydroxy-3,7-bis(2,4,6-tricyclohexylphenyl)-, 5-oxide, (11a<i>S</i>)- Solvents: Dichloromethane; -30 °C; 36 h, -30 °C Experimental Protocols	Enantioselective three-component Ugi reaction catalyzed by chiral phosphoric acid By: Zhang, Jian; et al Science China: Chemistry (2020), 63(1), 47-54.
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Scheme 34 (1 Reaction)

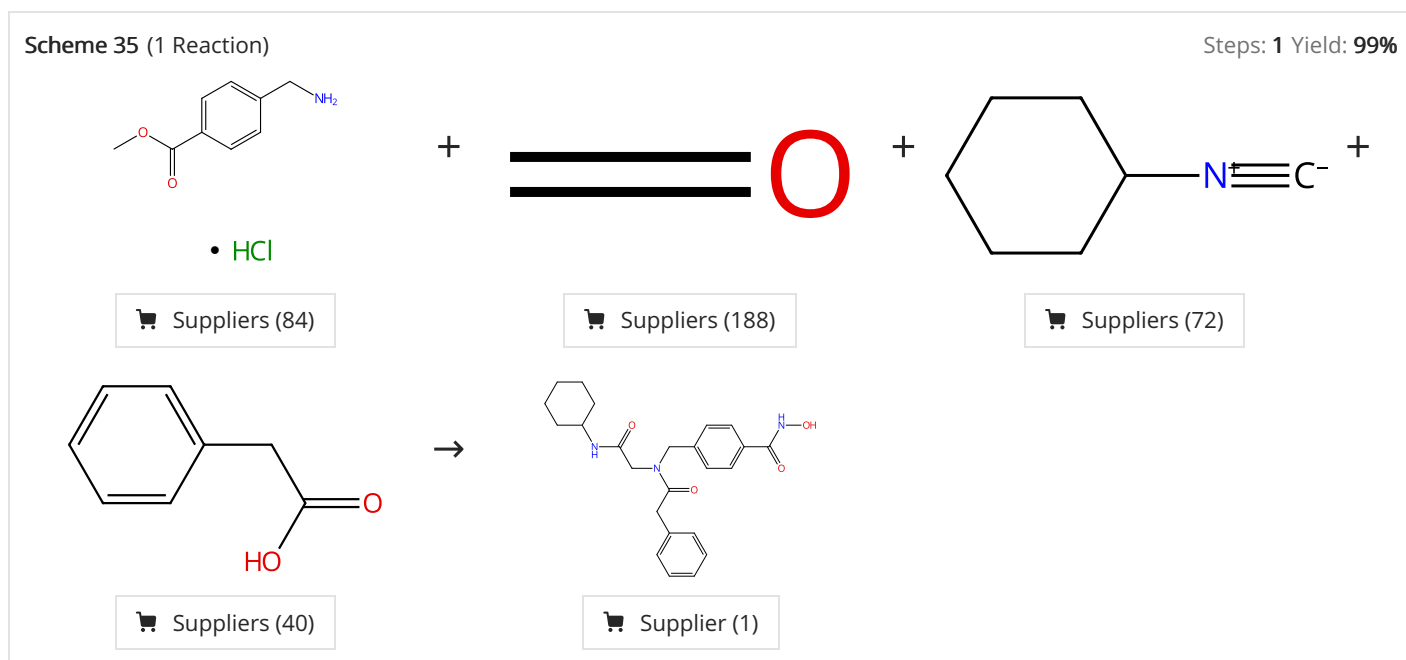
Steps: 1 Yield: 99%



Absolute stereochemistry shown, Rotation (-)

Suppliers (54)

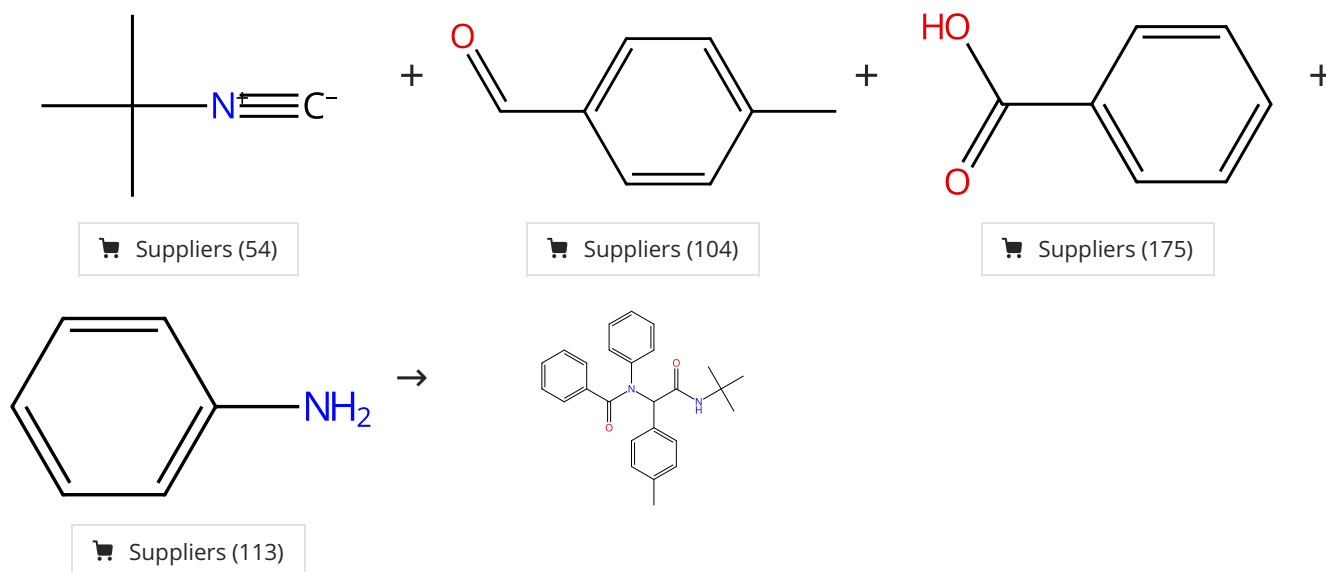
31-614-CAS-34981570 Steps: 1 Yield: 99% 1.1 Catalysts: Cobaltate(1-), bis[<i>N</i>-[[3,5-bis(1,1-dimethylethyl)-2-(hydroxy-κO)phenyl]methylene]-3-methyl-L-valinato(2-)-$\kappa N, \kappa O$]-, sodium (1:1), (<i>OC</i>-6-22'-<i>O</i>)- Solvents: Toluene; 30 min, rt 1.2 30 min, rt $\rightarrow$ -40 °C 1.3 24 h, -40 °C 1.4 Reagents: Triethylamine; -40 °C Experimental Protocols	Enantioselective Ugi and Ugi-azide reactions catalyzed by anionic stereogenic-at-cobalt(III) complexes By: Sun, Bing-Bing; et al Nature Communications (2022), 13(1), 7065.
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31-367-CAS-23174143 Steps: 1 Yield: 99% 1.1 Reagents: Triethylamine Solvents: Methanol; 30 min, 45 °C 1.2 120 min, 45 °C 1.3 Reagents: Sodium hydroxide, Hydroxylamine Solvents: Methanol, Water; 45 °C $\rightarrow$ 0 °C; 20 min, 0 °C; 0 °C $\rightarrow$ rt; 16 h, rt 1.4 Reagents: Hydrochloric acid Solvents: Water; pH 7 - 8, rt Experimental Protocols	Investigation of the in vitro and in vivo efficacy of peptoid-based HDAC inhibitors with dual-stage antiplasmodial activity By: Mackwitz, Marcel K. W.; et al European Journal of Medicinal Chemistry (2021), 211, 113065.
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Scheme 36 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-42417609

Steps: 1 Yield: 99%

Deuterated reagents in multicomponent reactions to afford deuterium-labeled products

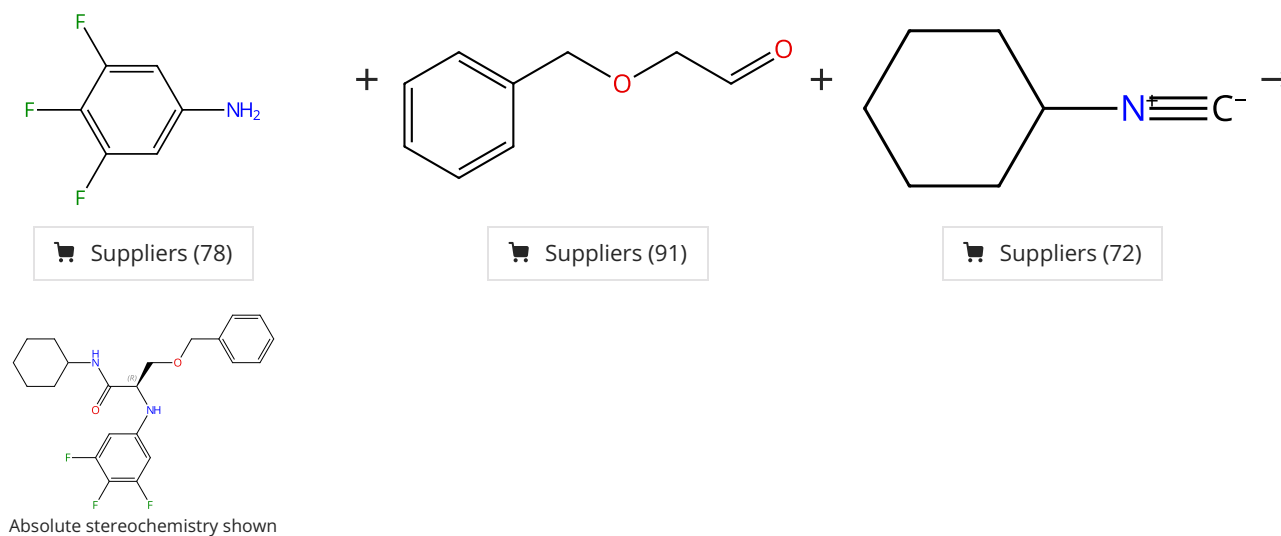
1.1 Solvents: Methanol; 24 h, rt

By: Schofield, Kevin; et al

Beilstein Journal of Organic Chemistry (2024), 20, 2270-2279.

Scheme 37 (1 Reaction)

Steps: 1 Yield: 98%



31-389-CAS-21608914

Steps: 1 Yield: 98%

Enantioselective three-component Ugi reaction catalyzed by chiral phosphoric acid

1.1 Catalysts: Diindeno[7,1-*de*:1',7'-*fg*][1,3,2]dioxaphosphocin, 10, 11,12,13-tetrahydro-5-hydroxy-3,7-bis(2,4,6-tricyclohexylphenyl)-, 5-oxide, (11a*S*)-

By: Zhang, Jian; et al

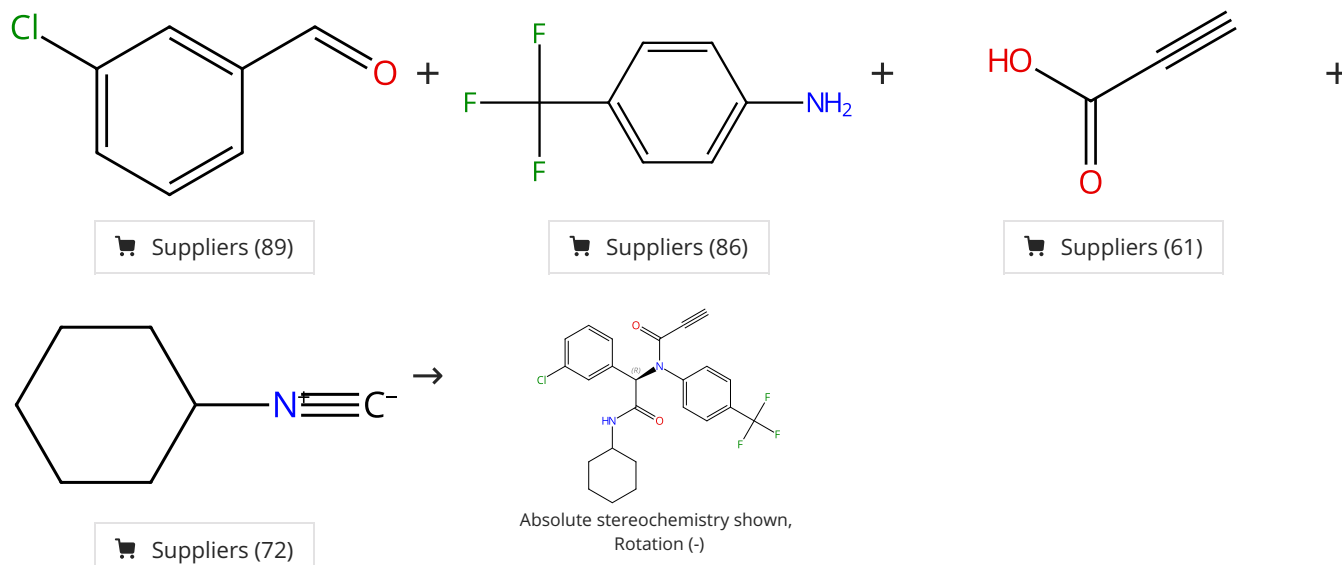
Solvents: Dichloromethane; -30 °C; 36 h, -30 °C

Science China: Chemistry (2020), 63(1), 47-54.

Experimental Protocols

Scheme 38 (1 Reaction)

Steps: 1 Yield: 98%



31-614-CAS-38204867

Steps: 1 Yield: 98%

- 1.1 **Catalysts:** Cobaltate(1-), bis[*N*-[[3,5-bis(1,1-dimethylethyl)-2-(hydroxy- κO)phenyl]methylene]-3-methyl-L-valinato(2-)- $\kappa N, \kappa O$]-, sodium (1:1), (*OC*-6-22'-*C*)-
Solvents: Toluene; 30 min, rt
 1.2 rt $\rightarrow$ -50 °C; 30 min, -50 °C
 1.3 12 h, -50 °C
 1.4 **Reagents:** Sodium bicarbonate
Solvents: Water

Experimental Protocols

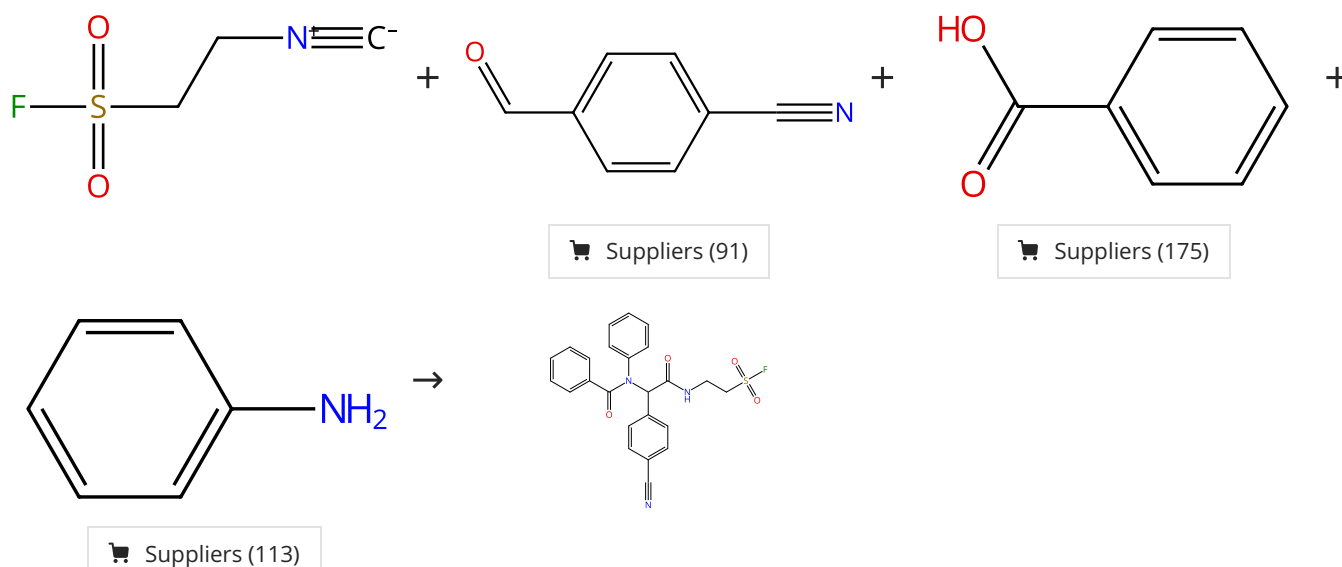
Enantioselective Access to Chiral 2,5-Diketopiperazines via Stereogenic-at-Cobalt(III)-Catalyzed Ugi-4CRs/Cyclization Sequences

By: Fang, Wei; et al

Journal of Organic Chemistry (2023), 88(22), 16024-16037.

Scheme 39 (1 Reaction)

Steps: 1 Yield: 98%



31-367-CAS-23902666

Steps: 1 Yield: 98%

- 1.1 **Solvents:** 1,1,1,3,3,3-Hexafluoro-2-propanol; 3 h, rt

Experimental Protocols

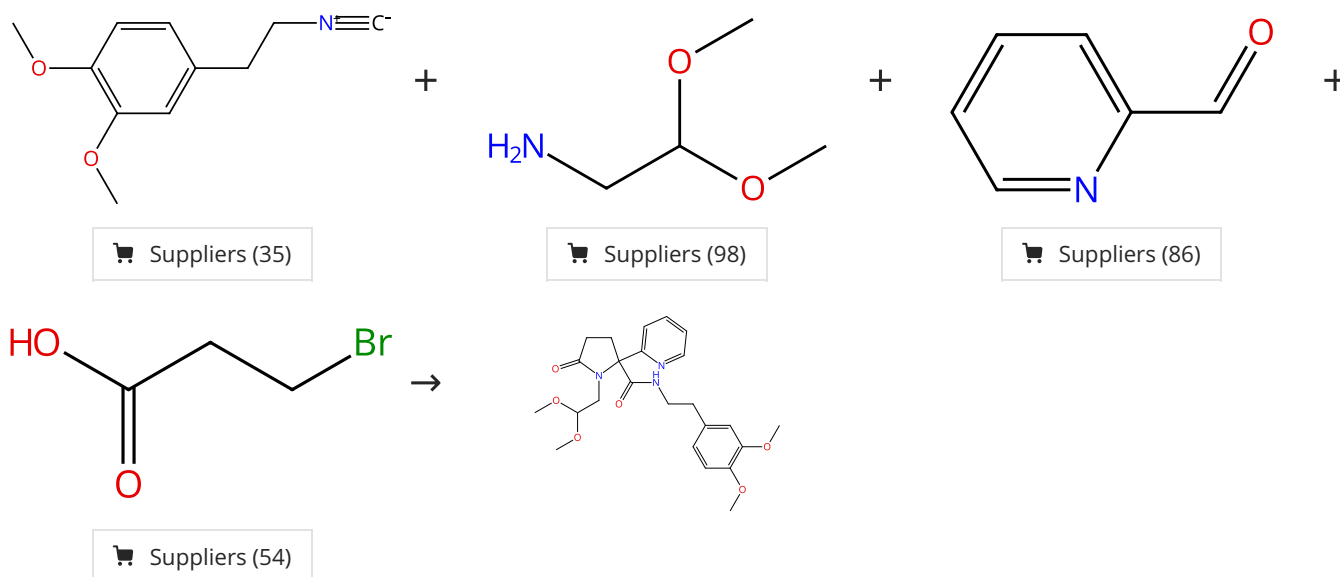
SuFExable isocyanides for Ugi reaction: Synthesis of sulfonyl fluoro peptides

By: Xu, Shuheng; et al

Organic Letters (2021), 23(13), 5197-5202.

Scheme 40 (1 Reaction)

Steps: 1 Yield: 98%



31-614-CAS-31999216

Steps: 1 Yield: 98%

Highly Stereoselective Ugi/Pictet-Spengler Sequence

1.1 Solvents: Methanol; 15 h, rt

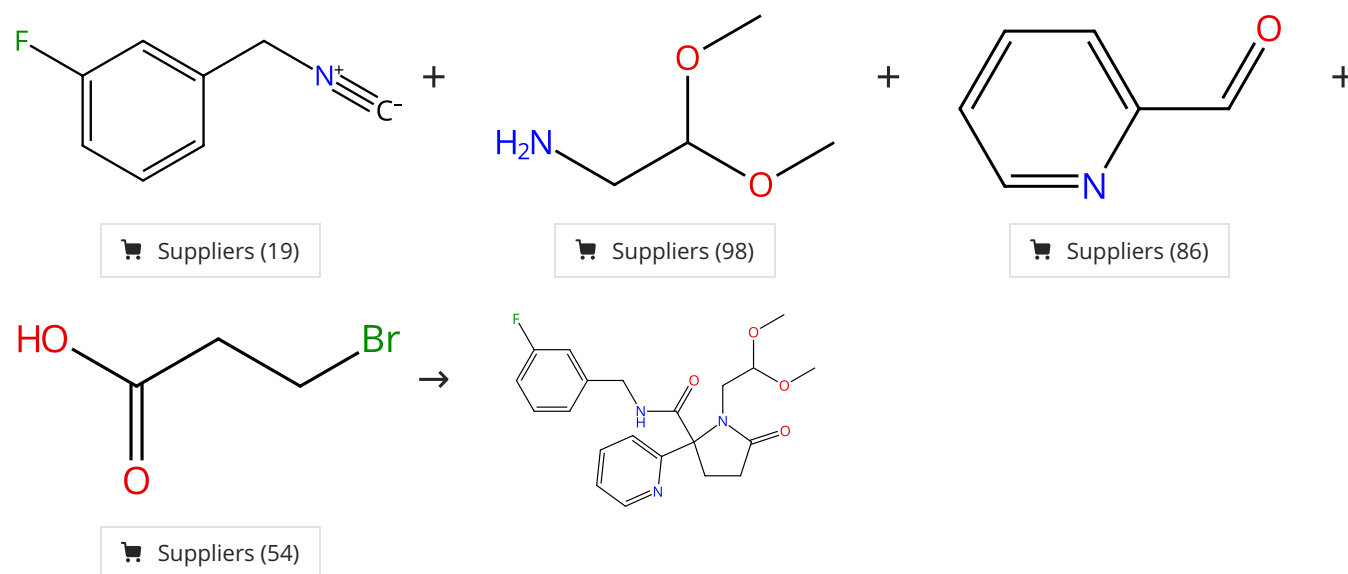
By: Zhang, Bidong; et al

Experimental Protocols

Journal of Organic Chemistry (2022), 87(11), 7085-7096.

Scheme 41 (1 Reaction)

Steps: 1 Yield: 98%



31-614-CAS-31999189

Steps: 1 Yield: 98%

Highly Stereoselective Ugi/Pictet-Spengler Sequence

1.1 Solvents: Methanol; 15 h, rt

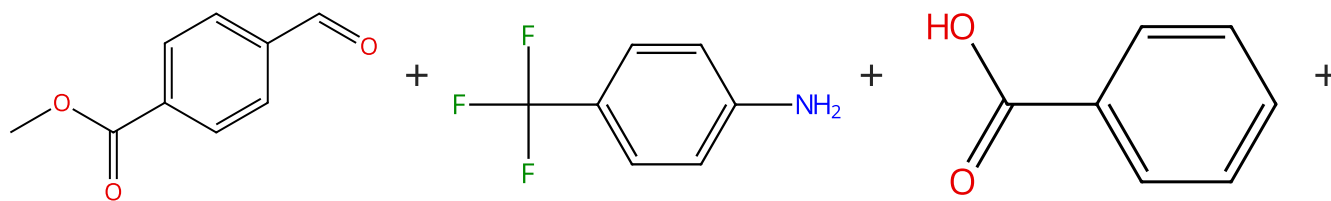
By: Zhang, Bidong; et al

Experimental Protocols

Journal of Organic Chemistry (2022), 87(11), 7085-7096.

Scheme 42 (1 Reaction)

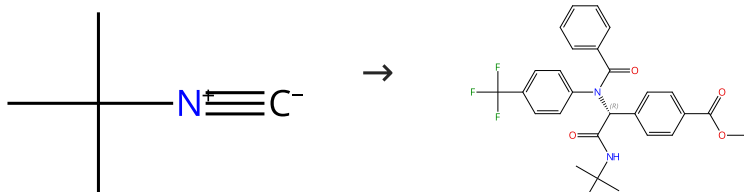
Steps: 1 Yield: 98%



Suppliers (99)

Suppliers (86)

Suppliers (175)



Suppliers (54)

Absolute stereochemistry shown,
Rotation (-)

31-614-CAS-34981567

Steps: 1 Yield: 98%

Enantioselective Ugi and Ugi-azide reactions catalyzed by anionic stereogenic-at-cobalt(III) complexes

By: Sun, Bing-Bing; et al

Nature Communications (2022), 13(1), 7065.

1.1 **Catalysts:** Cobaltate(1-), bis[*N*-[[3,5-bis(1,1-dimethylethyl)-2-(hydroxy-κ*O*)phenyl]methylene]-3-methyl-L-valinato(2-)-κ*N*, κ*O*]-, sodium (1:1), (*OC*-6-22'-*C*)-
Solvents: Toluene; 30 min, rt

1.2 30 min, rt → -40 °C

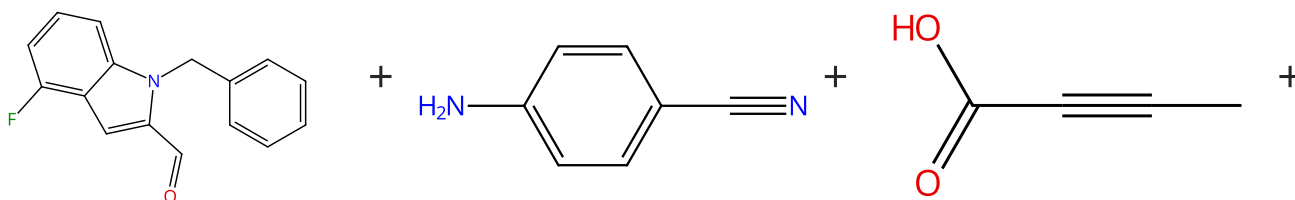
1.3 24 h, -40 °C

1.4 **Reagents:** Triethylamine; -40 °C

Experimental Protocols

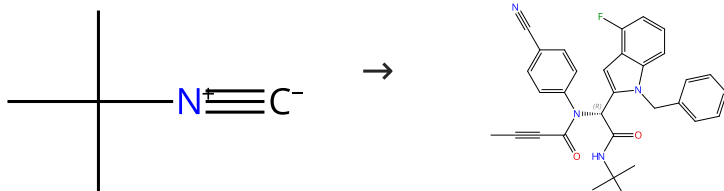
Scheme 43 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (100)

Suppliers (85)



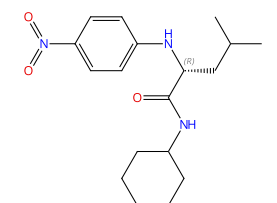
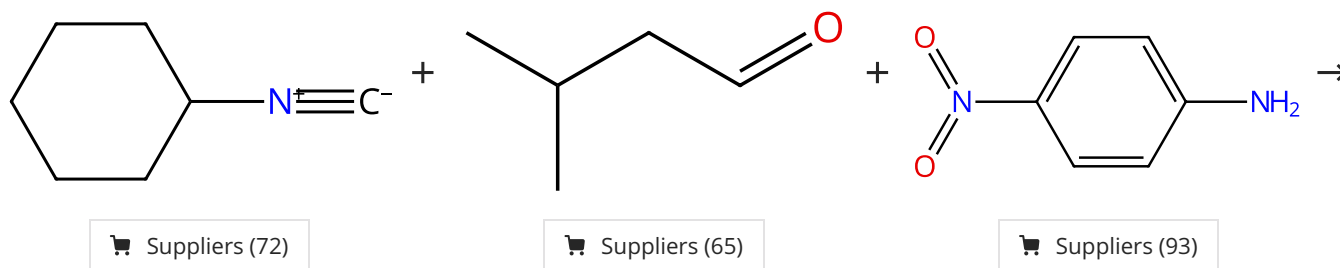
Suppliers (54)

Absolute stereochemistry shown,
Rotation (-)

31-614-CAS-45157635	Steps: 1 Yield: 98%	Asymmetric Synthesis of Azepinoindoles via Enantioselective Ugi-4CR/post-Ugi Transformation Sequence By: Fang, Li-Ping; et al Asian Journal of Organic Chemistry (2025), 14(6), e202500101.
1.1 Catalysts: Cobaltate(1-), bis[<i>N</i> -[[3,5-bis(1,1-dimethylethyl)-2-(hydroxy- κO)phenyl]methylene]-3-methyl-L-valinato(2-)- $\kappa N, \kappa O$]-, sodium (1:1), (<i>OC</i> -6-22'- <i>A</i>)- Solvents: Ethyl acetate; 30 min, rt		
1.2 30 min, -50 °C		
1.3 60 h, -50 °C		
1.4 Reagents: Sodium bicarbonate Solvents: Water; -50 °C		
Experimental Protocols		

Scheme 44 (1 Reaction)

Steps: 1 Yield: 98%

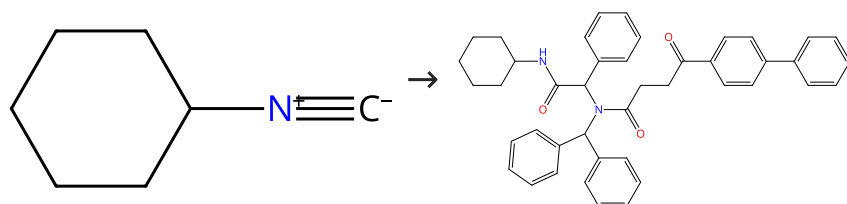
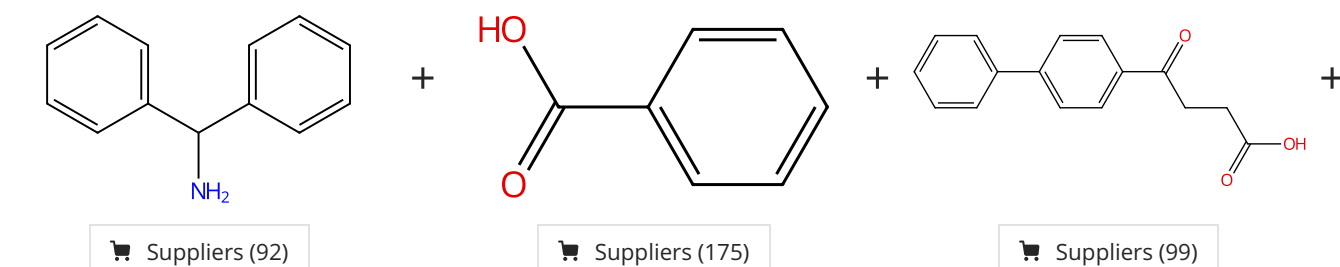


Absolute stereochemistry shown

31-389-CAS-21608904	Steps: 1 Yield: 98%	Enantioselective three-component Ugi reaction catalyzed by chiral phosphoric acid By: Zhang, Jian; et al Science China: Chemistry (2020), 63(1), 47-54.
1.1 Catalysts: Diindeno[7,1- <i>de</i> :1',7'- <i>fg</i>][1,3,2]dioxaphosphocin, 10, 11,12,13-tetrahydro-5-hydroxy-3,7-bis(2,4,6-tricyclohexylphenyl)-, 5-oxide, (11 <i>aS</i>)- Solvents: Dichloromethane; -30 °C; 36 h, -30 °C		
Experimental Protocols		

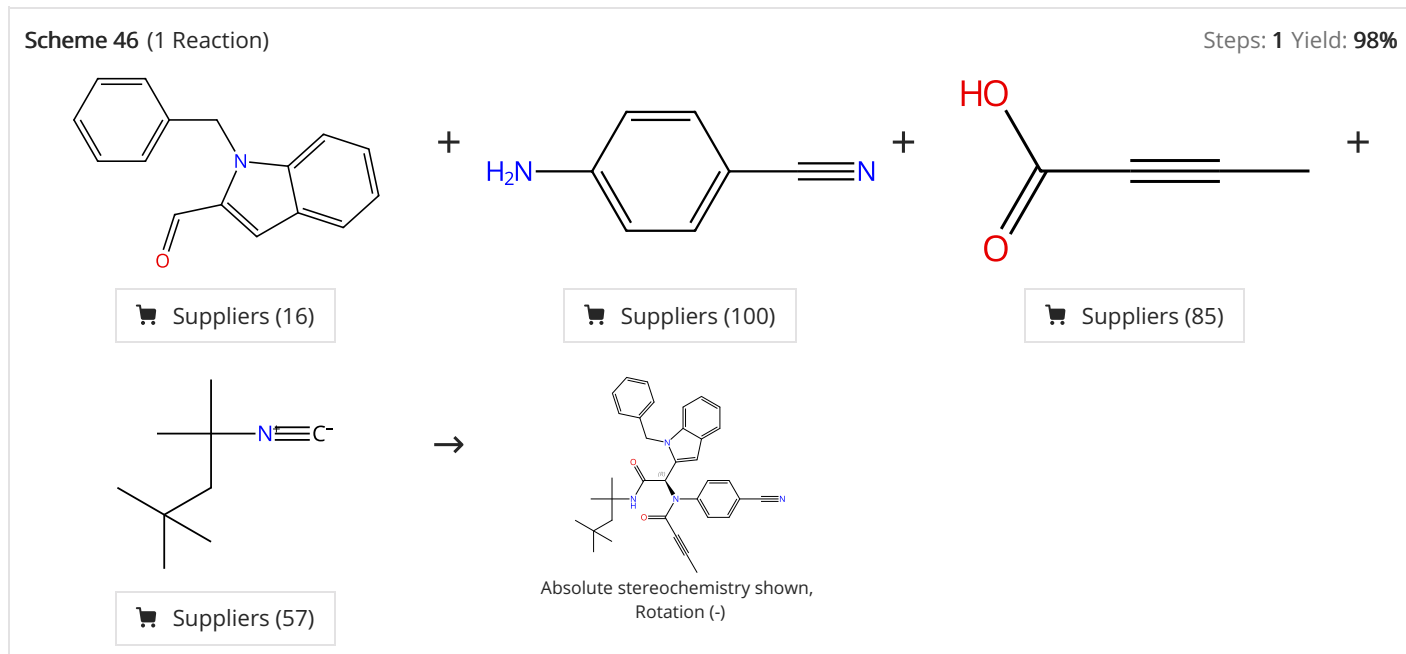
Scheme 45 (1 Reaction)

Steps: 1 Yield: 98%

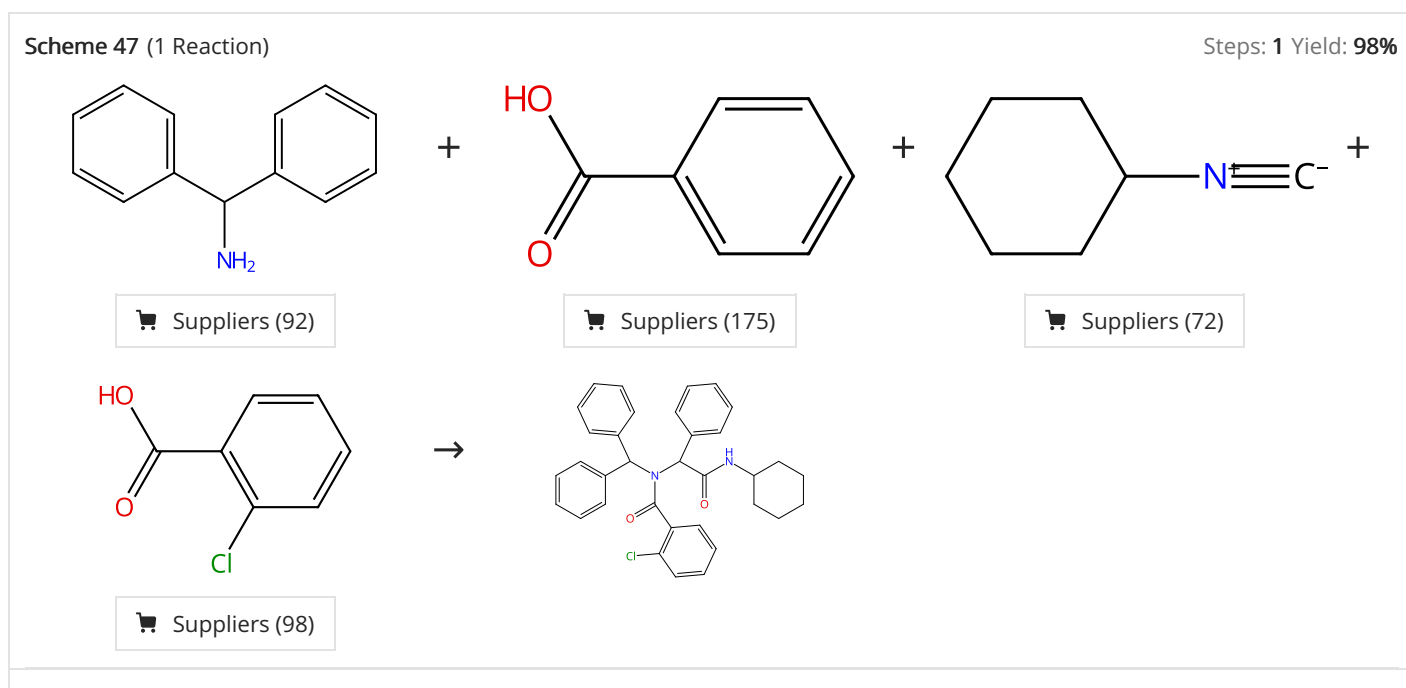


Suppliers (72)

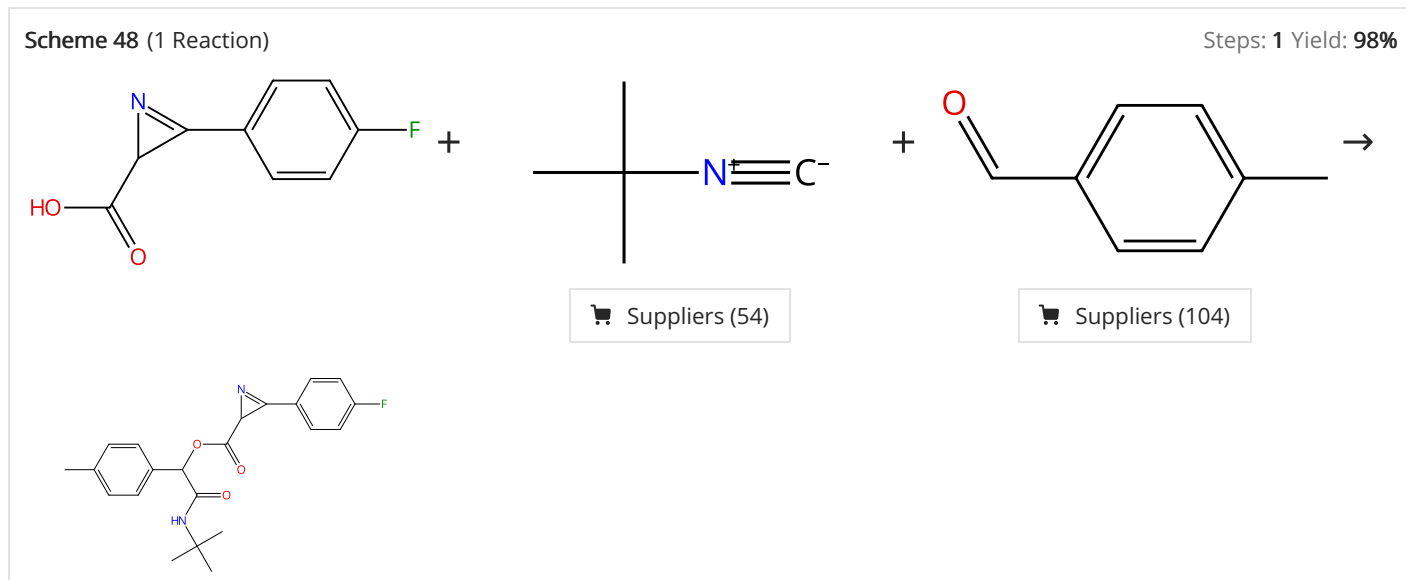
31-614-CAS-46080832 Steps: 1 Yield: 98%	A Reductive Ugi Reaction with Carboxylic Acids Replacing Aldehydes
1.1 Reagents: Methyldiethoxysilane Catalysts: Titanium tetrabutoxide; 5 h, 100 °C; 100 °C → rt	By: Li, Xiaohong; et al
1.2 Solvents: Methanol; 1 min, rt 1.3 8 h, rt	Organic Letters (2025), 27(22), 5899-5904.
Experimental Protocols	



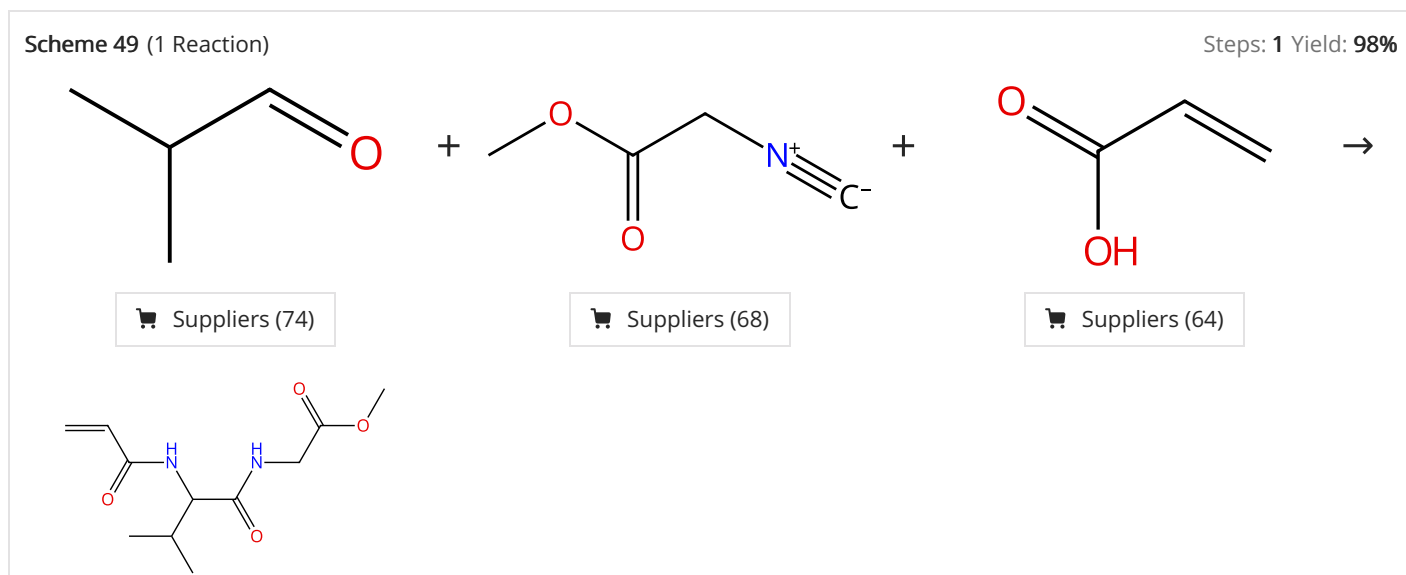
31-614-CAS-45157631 Steps: 1 Yield: 98%	Asymmetric Synthesis of Azepinoindoles via Enantioselective Ugi-4CR/post-Ugi Transformation Sequence
1.1 Catalysts: Cobaltate(1-), bis[<i>N</i> -[[3,5-bis(1,1-dimethylethyl)-2-(hydroxy- κ O)phenyl]methylene]-3-methyl-L-valinato(2-)- κ N, κ O]-, sodium (1:1), (<i>OC</i> -6-22'- <i>A</i>)-	By: Fang, Li-Ping; et al
Solvents: Ethyl acetate; 30 min, rt 1.2 30 min, -50 °C 1.3 60 h, -50 °C	Asian Journal of Organic Chemistry (2025), 14(6), e202500101.
1.4 Reagents: Sodium bicarbonate Solvents: Water; -50 °C	
Experimental Protocols	



31-614-CAS-46080779 Steps: 1 Yield: 98% 1.1 Reagents: Methyldiethoxysilane Catalysts: Titanium tetrabutoxide; 5 h, 100 °C; 100 °C → rt 1.2 Solvents: Methanol; 1 min, rt 1.3 8 h, rt Experimental Protocols	A Reductive Ugi Reaction with Carboxylic Acids Replacing Aldehydes By: Li, Xiaohong; et al Organic Letters (2025), 27(22), 5899-5904.
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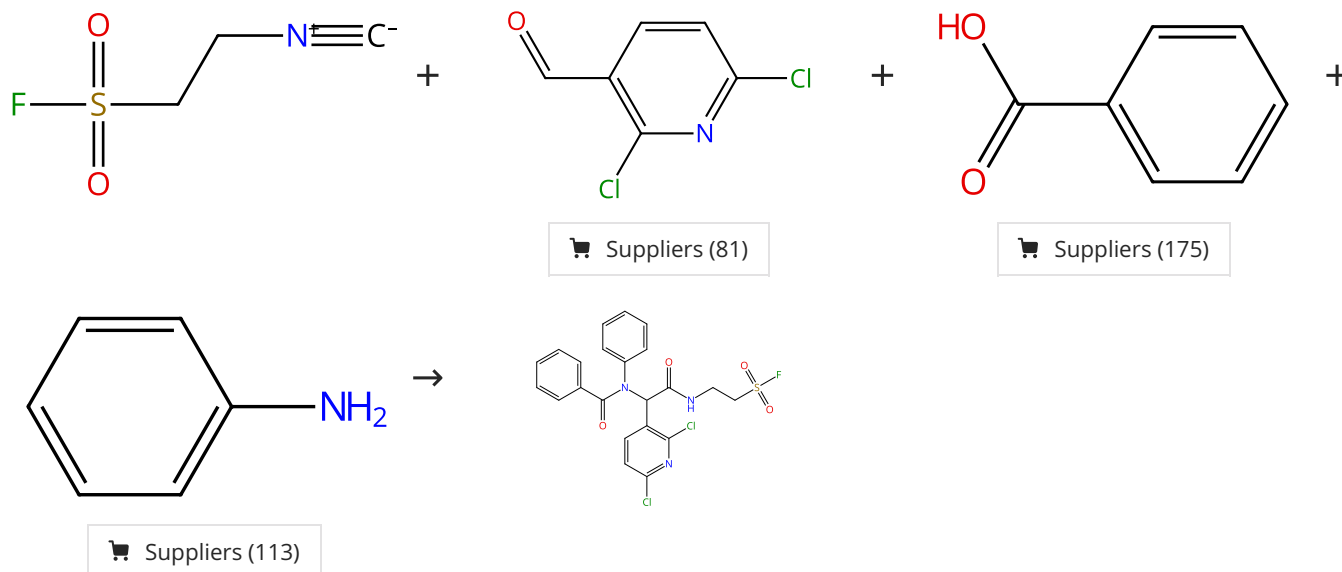
31-389-CAS-23036087 Steps: 1 Yield: 98% 1.1 24 - 48 h, rt	Azirine-containing dipeptides and depsipeptides: Synthesis, transformations and antibacterial activity By: Rostovskii, Nikolai V.; et al Organic & Biomolecular Chemistry (2020), 18(46), 9448-9460.
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31-614-CAS-34665209 Steps: 1 Yield: 98% 1.1 Reagents: Ammonium hydroxide Solvents: 2,2,2-Trifluoroethanol, Water; 15 min, rt 1.2 overnight, rt Experimental Protocols	Combining High-Throughput Synthesis and High-Throughput Protein Crystallography for Accelerated Hit Identification By: Sutanto, Fandi; et al Angewandte Chemie, International Edition (2021), 60(33), 18231-18239.
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Scheme 50 (1 Reaction)

Steps: 1 Yield: 98%



31-367-CAS-23903983

Steps: 1 Yield: 98%

SuFExable isocyanides for Ugi reaction: Synthesis of sulfonyl fluoro peptides

1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 3 h, rt

By: Xu, Shuheng; et al

Experimental Protocols

Organic Letters (2021), 23(13), 5197-5202.